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An Axiomatization of the Banzhaf Value

By E. Lehrer¹

Abstract: An axiomatization of the Banzhaf value is given. It is based on a version of three axioms, which are common to all the semi-values, and on an additional reduction axiom.

1 Introduction

A lot of effort had been devoted to the research of the semi values of games [2], especially to the Shapley-value. An axiomatization was given to the Shapley value [6], and to the Banzhaf value [3, 4].

The axiomatization given in [4] is based on the tool of the compound game and it does not determine the Banzhaf value uniquely. The axiomatization given in [3] treats a different approach. It is based on four axioms, three of them are standard and in fact determine the semi values [2]. We adopt here these three axioms.

We replace the fourth axiom of [3] by some reduction axiom, which has an intuitive meaning. This axiom says something about the value of a game in terms of another game, with smaller number of players.

It is worthwhile to mention that the use of such axioms is common in axiomatization of solution concepts (see for example: in [5] an axiomatization of the prekernel, and in [7] an axiomatization of the prenucleolus).

We define here for each coalition T two T -games derived from the original game by amalgamting all the players of T to one player called $\bar{T}$. Our additional axiom says that for any two-players-coalition $T = \{i, j\}$ the sum of the values of i and of j in the original game is less or equal to the value of $\bar{T}$ in the T -games. According to this axiom a unification of any two players is profitable.

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The main significance of the following axiomatization is the indication of the difference between the Banzhaf value and the Shapley value. For example take the three player weighted majority game $\left[1; \frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right]$; its Banzhaf value is $\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$ while its Shapley value is $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$. After amalgamating players 1 and 2 to a new player we will get the game $\left[1; \frac{2}{3}, \frac{1}{3}\right] = \left[1; \frac{1}{2}, \frac{1}{2}\right]$. The Banzhaf value and the Shapley value coincide on the two player games and they are equal in this particular game to $\left(\frac{1}{2}, \frac{1}{2}\right)$. Notice that $\frac{1}{4} + \frac{1}{4} \leq \frac{1}{2}$, while $\frac{1}{3} + \frac{1}{3} > \frac{1}{2}$. In other words, in this case according to the Shapley value unification is harmful while according to the Banzhaf value unification is never harmful, and in some cases it is profitable.

2 Definitions, the Axiom Systems, and Theorems

Before introducing the axioms we need several notations. A *game in coalitional function form* is a pair (N, v) where N is a finite set of players and v is a real valued function defined on the subsets of N and $v(\emptyset) = 0$. A *simple game* is a game (N, v) where v is ranged to $\{0, 1\}$. In a simple game all the coalitions $S \subset N$ with $v(S) = 1$ are called *winning* and all the rest *losing*.

A *monotonic game* is a game (N, v) which satisfies the condition whereby $S \subseteq T$ implies $v(S) \leq v(T)$.

Let $T \subseteq N$, $|N| = n$ the *T-unanimity game* denoted by $U_T^n = (N, u_T^n)$ is defined to be $u_T^n(S) = 1$ if $T \subseteq S$ and otherwise 0. Whenever $|T| = 1$ U_T^n is called *dictatorial*. Let $k \leq |T|$, the *k-T symmetric game* denoted by $S_T^{k,n} = (N, s_T^k)$ is defined to be $s_T^k(A) = 1$ if $k \leq |T \cap A|$ and otherwise 0.

We denote by G , SG and MSG the sets of all games, all the simple games and all the monotonic simple games respectively.

In order to simplify let us denote the game (N, v) by v whenever ambiguity can not arise.

Let v and u be two simple games. Define the games $v \vee u$ and $v \wedge u$ as follows:
For any coalition S

$$v \vee u(S) = \text{Max} \{v(S), u(S)\}$$

$$\text{and } v \wedge u(S) = \text{min} \{v(S), u(S)\}. \quad (1)$$

The Banzhaf value η corresponds to each $(N, v) \in G$ a vector $\eta(v)$ in $\mathbb{R}^{|N|}$. $\eta(v)$ is defined as follows:

$$\eta^i(v) = \frac{1}{2^{|N|-1}} \sum_{S \subseteq N} v(S \cup \{i\}) - v(S) \quad 1 \leq i \leq n.$$

If v is an n players monotonic simple game, $2^{n-1} \eta^i(v)$ counts the number of losing coalitions S that become winning after i joins them. Here we arrive at the definitions of the T -games. Let $(N, v) \in G$ and $T \subseteq N$. In case all the players in T are amalgamated into one player $\bar{T}$, two games can be derived:

$$(N - T \cup \{\bar{T}\}, v_T) \quad \text{and} \quad (N - T \cup \{\bar{T}\}, v_T^m),$$

where for each $S \subseteq N - T$

$$v_T(S) = v(S), \quad v_T(S \cup \{\bar{T}\}) = v(S \cup T)$$

and² (2)

$$v_T^m(S) = v(S), \quad v_T^m(S \cup \{\bar{T}\}) = \max_{\phi \neq B \subseteq T} v(S \cup B).$$

Notice that whenever v is monotonic, $v_T = v_T^m$.

We will use two axiom-systems, one for SG solely and the other for G . The two systems differ from one another only by the linearity³ axiom used.

The first system is:

(D) If $v(S \cup \{i\}) = v(S) + v(\{i\})$ for every coalition S such that $i \notin S$ then $\varphi^i(v) = v(\{i\})$.

(ET) If for every coalition $S \subseteq N \setminus \{i, j\}$ $v(S \cup \{i\}) = v(S \cup \{j\})$ then $\varphi^i(v) = \varphi^j(v)$.

(UI)⁴ $\varphi(v \vee u) + \varphi(v \wedge u) = \varphi(v) + \varphi(u)$.

² The m in v_T^m stands for max.

³ We call it linearity because additivity (or super additivity) is reserved for the fourth axiom.

⁴ For union-intersection.

(SA) $\varphi^i(v) + \varphi^j(v) \leq \varphi^{\bar{T}}(v_T)$ for every two-players-coalition $T = \{i, j\}$.

(SA^m) $\varphi^i(v) + \varphi^j(v) \leq \varphi^{\bar{T}}(v_T^m)$ for every two-players-coalition $T = \{i, j\}$.

(D) is the dummy axiom used by Shapley [6].

(ET) is the equal treatment axiom. It says that whenever v remains unchanged after interchanging i by j , $\varphi^i(v)$ equals to $\varphi^j(v)$.

(UI) is an axiom due to Dubey [1] and serves as the linearity axiom when the discussion concentrates on SG . Reasoning for (UI) can be found in [3]⁵.

If a question arises as for the amalgamation of coalitions containing more than two players, one can argue that the amalgamation of "big" coalitions (according to (SA)) is done step by step. First two players are amalgamated to one player and then a third player is amalgamated to the new one, and so on. A technical answer could be given as follows: a general super additivity, namely $\sum_{i \in T} \varphi^i(v) \leq \varphi^{\bar{T}}(v_T)$ for all $T \subseteq N$, is not consistent with the other three axioms.

Theorem A: φ satisfies (D), (ET), (UI) and either (SA) or (SA^m) for every $v, u \in SG$ if and only if φ is the Banzhaf value.

In fact the following holds.

Proposition 1:

(i) If $v \in SG$ then

$$\eta^i(v) + \eta^j(v) \leq \eta^{\bar{T}}(v_T^m)$$

for every coalition $T = \{i, j\}$ and equality holds for every $T = \{i, j\}$ if and only if $v \in MSG$.

⁵ The fourth axiom of [3] is $\sum_{i \in N} \varphi^i(v) = \sum_{i \in N} \eta^i(v)$.

(ii) If $v \in SG$ then

$$\eta^i(v) + \eta^j(v) = \eta^{\bar{T}}(v_T)$$

for every coalition $T = \{i, j\}$.

It means that (SA) together with the other axioms give additivity instead of super additivity.

Remarks:

1. The Banzhaf value satisfies (UI) on SG as it will be shown below, while the Shapley value satisfies it on MSG and not on SG .
2. The axiom (SA) distinguishes between the Shapley value and the Banzhaf value. In terms of v_T the efficiency axiom can be written as

$$\sum_{i \in T} \varphi^i(v) = \varphi^{\bar{T}}(v_T) \text{ for every } |T| = |N|,$$

while Banzhaf value satisfies it for every $|T| = 2$.

3. Given that φ is equal to the Banzhaf value in all the two players games (which coincide with the Shapley value), the "2-efficiency" axiom:

$$\sum_{i \in T} \varphi^i(v) = \varphi^{\bar{T}}(v_T) \text{ for every } |T| = 2$$

determines the Banzhaf value uniquely (the proof appears in the Appendix).

Finally, we can use the standard linearity axiom of Shapley in order to axiomize the Banzhaf values on G .

$$(LI) \quad \varphi(v) + \varphi(u) = \varphi(u + v).$$

Theorem B: φ satisfies (D), (ET), (LI) and either (SA) or (SA^m) for every $v, u \in G$ if and only if φ is the Banzhaf value on G .

3 Proofs

Proposition 1 Proof: Let $(N, v) \in SG$, where $|N| = n$, then by the definition of η

$$\begin{aligned}
 2^{n-1}(\eta^i(v) + \eta^j(v)) &= \sum_{S \subseteq N} [v(S \cup \{i\}) - v(S)] + \sum_{S \subseteq N} [v(S \cup \{j\}) \\
 &\quad - v(S)] = \sum_{S \subseteq N \setminus \{i, j\}} [v(S \cup \{i\}) - v(S) + v(S \cup \{i, j\}) - v(S \cup \{j\})] \\
 &\quad + \sum_{S \subseteq N \setminus \{i, j\}} [v(S \cup \{j\}) - v(S) + v(S \cup \{i, j\}) - v(S \cup \{i\})] \\
 &= 2 \sum_{S \subseteq N \setminus \{i, j\}} [v(S \cup \{i, j\}) - v(S)]
 \end{aligned}$$

The last term is equal to $2 \cdot 2^{n-2} \eta^{\bar{T}}(v_T)$ for $T = \{i, j\}$ (this gives the proof for (ii)) and furthermore it is less or equal to

$$2 \cdot 2^{n-1} \eta^{\bar{T}}(v_T^m) = 2 \sum_{\phi \neq B \subseteq \{i, j\}} \text{Max } v(S \cup B) - v(S).$$

$v \in MSG$ if and only if for every $\{i, j\}$ and $S \subseteq N \setminus \{i, j\}$

$$v(S \cup \{i, j\}) = \text{Max}_{\phi \neq B \subseteq \{i, j\}} v(S \cup B).$$

So equality holds iff $v \in MSG$ as desired at (i).

Theorems A and B Proofs: It is known that η satisfies (D), (ET) and (LI). Proposition 1 shows that η satisfies also (SA) and (SA^m). Thus in order to show that η satisfies all the axioms it remains to show that for every $v, u \in SG$

$$\varphi(v \vee u) + \varphi(v \wedge u) = \varphi(v) + \varphi(u).$$

Let $S \subseteq N$ and $i \in N$

$$\begin{aligned}
 2^{n-1}(\eta^i(v \vee u) + \eta^i(v \wedge u)) &= \sum_{S \subseteq N} \text{Max } \{v(S \cup i), u(S \cup i)\} \\
 &\quad - \text{Max } \{v(S), u(S)\} + \sum_{S \subseteq N} \text{Min } \{v(S \cup i), u(S \cup i)\}
 \end{aligned}$$

$$\begin{aligned}
 -\text{Min } \{v(S), u(S)\} &= \sum_{S \subseteq N} [v(S \cup i) + u(S \cup i) - v(S) - u(S)] \\
 &= 2^{n-1}(\eta^i(v) + \eta^i(u)).
 \end{aligned}$$

The first step in order to prove the other direction of Theorem A will be to show that the axiom system determines the values of φ on the unanimity games. This step is proved through two inductions simultaneously, the first one is on the players number and the second is on the number of the non-dummy players in the unanimity game. We give here the induction step.

We assume that φ is already determined on the simple games with n players and on the games of the type U_T^{n+1} where $|T| \leq k$. Furthermore, we assume that

$$\varphi^i(U_T^{n+1}) = 1/2^{|T|-1} \text{ if } i \in T \text{ and otherwise } 0.$$

Let $T \subseteq \{1, \dots, n+1\}$ where $|T| = k+1$, and $i \in T$. Denote

$$T' = T \setminus \{i\}.$$

By (UI):

$$\varphi(U_T^{n+1} \vee U_{\{i\}}^{n+1}) + \varphi(U_T^{n+1} \wedge U_{\{i\}}^{n+1}) = \varphi(U_{T'}^{n+1}) + \varphi(U_{\{i\}}^{n+1}). \quad (3)$$

By (D) and (ET) there are constants a , b and c s.t.

$$\varphi^j(U_T^{n+1}) = \begin{cases} a & j \in T \\ 0 & j \notin T \end{cases} \quad (4a)$$

and

$$\varphi^j(U_T^{n+1} \vee U_{\{i\}}^{n+1}) = \begin{cases} b & j \in T' \\ c & j = i \\ 0 & \text{otherwise.} \end{cases} \quad (4b)$$

By the induction hypothesis and by (4a), (4b)

$$a + b = 1/2^{k-1} \quad (5a)$$

and with (D) we get

$$a + c = 1. \quad (5b)$$

Now, in order to apply (SA) (or (SA^m)) take two players i, j from T and amalgamate them to one player. The game which derives is $U_{T'}^n$ (after identifying $j \in T'$ with the new player $\bar{ij}$).

Hence by (SA) (or by (SA^m))

$$2a \leq 1/2^{|T'|-1} = 1/2^{k-1}. \quad (6a)$$

Secondly, amalgamate the player i with any player from T' in the game $U_{T'}^{n+1} \vee U_{\{i\}}^{n+1}$. The game which derives is a dictatorial game so,

$$b + c \leq 1. \quad (6b)$$

Add (6a) to (6b) to get

$$2a + b + c \leq 1 + \frac{1}{2^{k-1}} \quad (7)$$

(5a), (5b) and (7) bring us to conclude that the inequalities in (6a) and (6b) become equalities. In particular

$$2a = \frac{1}{2^{k-1}}.$$

This gives the value of φ on U_T^{n+1} .

Theorem B's proof has the same procedure and we'll give here the induction step:
Let $|T| = k + 1$ by (LI)

$$\varphi((|T| - 1)U_T^{n+1}) + \varphi(S_T^{k,n+1}) = \sum_{\substack{T' \subseteq T \\ |T'| = k}} \varphi(U_{T'}^{n+1}) \quad (8)$$

By (ET) and (D) there are constants a and b s.t.

$$\varphi^i(U_T^{n+1}) = \begin{cases} a & i \in T \\ 0 & i \notin T \end{cases}$$

and

$$\varphi^i(S_T^{k,n+1}) = \begin{cases} b & i \in T \\ 0 & i \notin T. \end{cases}$$

By (8) and by the induction hypothesis

$$ka + b = (|T| - 1)/2^{k-1} = k/2^{k-1}. \quad (9)$$

Amalgamate i to j ($i, j \in T$) in U_T^{n+1} and derive the game $U_{T'}^n$ (as above $T' = T \setminus \{i\}$).

Hence by (SA) (or (SA^m))

$$2a \leq \frac{1}{2^{k-1}}. \quad (10a)$$

Similarly by amalgamating i to j in $S_T^{k,n+1}$ (for some i and j in T) we will derive the game

$((N \setminus \{i, j\}) \cup \{\bar{i}\bar{j}\}, v')$ where

$$v'(A) = \begin{cases} 1 & \bar{i}\bar{j} \in A \text{ and } k-2 \leq |(A \setminus \{\bar{i}\bar{j}\}) \cap T| \\ 0 & \text{otherwise.} \end{cases}$$

By the induction hypothesis (v' is a game with a smaller number of players),

$$\begin{aligned}\varphi^{ij}(v') &= k/2^{k-1}, \quad \text{thus} \\ 2b &\leq k/2^{k-1}.\end{aligned}\tag{10b}$$

From (10a) and (10b) we get

$$ka + b \leq k/2^k + k/2^k = k/2^{k-1}\tag{11}$$

(9) and (11) give equalities in (10a) and in (10b) instead of inequalities. In particular $a = 1/2^k$ as desired. By the same way $\varphi^i(c \cdot U_T^n) = c/2^{|T|-1}$ whenever $i \in T$ and otherwise 0, for any $c > 0$ and $T \subseteq N$.

From now on, both proofs are matters of linearity. As shown in [3] (UI) determine together with the value of φ on the unanimity games the values of φ on *MSG*. To conclude the proof of Theorem A it will be noticed first that a simple game (N, v) can be represented as $\bigvee_{v(T)=1} W_T^n$ where $W_T^n(S) = 1$ only when $S = T$, and second that by (UI)

$$\begin{aligned}\varphi(U_T^n) &= \varphi(U_T^n - W_T^n \vee W_T^n) = \varphi(U_T^n - W_T^n \vee W_T^n) + \varphi(U_T^n - W_T^n \wedge W_T^n) \\ &= \varphi(U_T^n - W_T^n) + \varphi(W_T^n)\end{aligned}$$

so,

$$\varphi(W_T^n) = \varphi(U_T^n) - \varphi(U_T^n - W_T^n).$$

$\varphi(U_T^n)$ and $\varphi(U_T^n - W_T^n)$ are already determined because U_T^n and $U_T^n - W_T^n$ are both in *MSG*.

From the point in which φ is determined on the set $\{c \cdot U_T^{|N|} \mid c > 0, T \subseteq N\}$ on, the proof of Theorem B is essentially standard (see for example [4]).

Appendix

Proof of Remark 3: The proof is through induction on the players number. By the assumption φ is already determined on the two-players-games. Let $V = (\{1\}, v)$ a game with one player, then $v(1) = c$, to get $\varphi(v)$ take the following two players game u : $u(S) = c$ when $S = \{1, 2\}$ and 0 otherwise. After amalgamating 1 to 2 in u we will get v . By the assumption, $\varphi(u)$ is known ($\varphi(u) = \eta(u)$) so $\varphi^1(v) = \eta^1(u) + \eta^2(u)$ by “2-efficiency” axiom.

In order to establish that the Banzhaf value is determined uniquely on all the games, assume inductively that the Banzhaf value is already determined on all the games with n players or less. Take a game $V = (\{1, \dots, n+1\}, v)$ of $n+1$ players. By amalgamating i and j in V to a new player $\bar{ij}$ we will get the game $V_{\{i,j\}}$, which is of n players. By “2-efficiency” axiom we get $x_i + x_j = \varphi^{\bar{ij}}(V_{\{i,j\}})$ where $x_i = \varphi^i(V)$. We have got $\binom{n+1}{2}$ equations with $n+1$ variables; an equation for each pair $i, j \in \{1, \dots, n+1\}$. Obviously this equations system has degree $n+1$. It is consistent because $\varphi^{\bar{ij}}(V_{\{i,j\}}) = \eta^i(V) + \eta^j(V)$, by Proposition 1, and by the induction hypothesis. Thus it has a unique solution which is $x_i = \eta^i(V)$.

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The Manipulability of the Shapley-Value¹

By W. Thomson²

Abstract: The manipulation of the Shapley-value, when used as a resource allocation mechanism, is examined. First, the extent to which an individual can, by unilaterally misrepresenting his utility function, affect the value allocation in his favor, is evaluated. When all agents attempt to manipulate, a game results, whose equilibrium allocations can be described as follows. At an equilibrium, the initial allocation appears to be Pareto-efficient. Any equilibrium allocation is also an equilibrium allocation of the analogously defined Walrasian manipulation game. The true (constrained) Walrasian allocations are equilibrium allocations. Under two slight respecifications of the value, there are no other equilibrium allocations.

1 Introduction

1.1 The Problem

The *Shapley-value*, or *value* (Shapley 1953), is a game theoretic solution concept that has been applied to the economic problem of fairly dividing a bundle of commodities among a group of agents (Shapley and Shubik 1969). The determination of *value allocations* requires the knowledge of the agents' utility functions, which are unfortunately not directly observable; one has to rely on what the agents claim them to be. It is usually the case, however, that an agent can benefit from misrepresentation of his utility function. If all agents engage in such manipulation, the resulting outcomes will be different from the ones to which truthful behavior would have led. The object of

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this paper is to describe the consequences of manipulative behavior, in terms of efficiency and distribution. We will provide a characterization of the equilibrium allocations of the "manipulation game" associated in a natural way with the value.

The approach followed here should be contrasted with that followed in most of the literature on incentive-compatibility; the focus there mainly has been on the characterization of the class of economies over which manipulation-proof mechanisms can be defined, and on the construction of such mechanisms. Far fewer studies have been devoted to the descriptive study of the manipulability of known economic mechanisms. Once it has been established that a mechanism is not immune to manipulation, it remains to identify what outcomes are likely to result from manipulation. We feel that the extent to which the satisfactory operation of an economic mechanism is disrupted by manipulative behavior, or its *manipulability*, should enter in an important way in the overall evaluation of the mechanism.

1.2 Literature Review

The contributions in which similar questions have been addressed can conveniently be divided into two groups, depending on whether the manipulation game was found to have equilibria in dominant strategies or in Nash strategies. Kannai (1977) considers the manipulation game played by two agents whose ordinal preferences, assumed to be concavifiable and defined over an l -dimensional commodity space are known and the outcome is determined according to the Nash bargaining solution, and shows that the least concave representation of his preferences is a dominant strategy for each agent. Kilhstrom, Roth, and Schmeidler (1979), and Roth (1979), analyze a class of two-person bargaining solutions including the Nash (1950) and the Kalai and Smorodinsky (1975) solutions. They find that, for a large class of games, a player would always prefer facing a more, rather than a less, risk-averse opponent. This result implies that linear utilities are dominant strategies in the manipulation game associated with the Nash solution when applied to the one-commodity division problem, as directly established by Crawford and Varian (1979). Kurz (1977) proves that linear utilities are dominant strategies also in the manipulation game associated with the Aumann and Kurz (1977a) one-commodity redistribution model based on the Nash solution and the value, in economies with a continuum of agents; in the multi-commodity version (Aumann and Kurz 1977b), Kurz (1980) finds dominant strategies that are functions homogeneous of degree one. Mas-Colell (1980) derives similar results for the manipulation of the value in a production economy with an infinite number of agents falling into a finite number of types.

In the second group of contributions, equilibria are found to exist only in Nash-equilibria. First, we mention the work of Hurwicz (1978) and Otani and Sicilian

(1982), whose results, although not directly pertaining to the value, will be helpful in describing and understanding ours. Hurwicz studies the manipulation game associated with the Walrasian mechanism in two-commodity, two-person economies, and characterizes its set of equilibrium allocations as the interior of the lens-shaped area bounded by the true offer curves. This characterization is extended by Otani and Sicilian to the l -commodity, two-person case. We close this review with a statement of Sobel's (1981) main results, which are the most closely related to ours. Sobel studies the application to l -commodity fair division problems to a class of two-person bargaining solutions containing the Nash and Kalai-Smorodinsky solutions. He finds that the true equal income (constrained)³ Walrasian allocations are always equilibrium allocations of their associated manipulation games and the only equilibrium allocations when agents are required to announce linear utilities. Also, equilibrium allocations are envy-free for the true preferences. Finally, for those two solutions, any equilibrium allocation is an equal income (constrained) Walrasian allocation for the reported utilities and any equilibrium allocation that is efficient for the true utilities is a true equal income (constrained) Walrasian allocation.

1.3 Summary of the Results

The subject of the present paper is the manipulability of the value in transferable utility economies.⁴

We start out by assuming that all but one agent are honest, and we study what this last agent can achieve through unilateral manipulation. This analysis parallels the traditional analysis of the vulnerability to unilateral manipulation of the Walrasian mechanism, and the well-known characterization of the "monopoly point", the allocation obtained by the monopolist under optimal manipulation. Our first result is that the counterpart of this allocation for the value, that is, the allocation obtained by the cheater under unilateral and optimal manipulation of the value, is always preferred by him, according to his true preferences, to the monopoly allocation. Of course, this result does not imply that the value is "more manipulable" than the Walrasian mechanism, since there are other natural criteria on which to base such

³ The difference between the *constrained* Walrasian correspondence, as opposed to the usual Walrasian correspondence, mainly is a technical one, which we discuss in greater detail later (Section 6.2).

⁴ Shapley (1969) proposed an extension of the value to economies without transferable utility, the NTU value. Recently, this extension has been the object of a large amount of attention. The manipulation of the NTU value is not examined here.

comparisons. First, as we show next, the *gain* achieved by unilateral and optimal manipulation of the value as compared to honest behavior may be greater, but it may also be smaller, than the corresponding gain for the Walrasian mechanism. (Note that gains can be meaningfully measured, and compared, in terms of the commodity that permits utility transfers.) Second, it is more natural to assume the same degree of sophistication on the part of all agents. What happens when they *all* attempt to manipulate?

To answer this question, we study the value *manipulation quasi-game*⁵, the game in which strategies are utility functions and the outcome correspondence is the correspondence associating with the economy defined by each list of strategies its set of value allocations. It is to the study of this quasi-game that most of the paper is devoted.

The most important conclusion of our study concerns the central role played by the Walrasian correspondence in the determination of the equilibrium allocations of the value quasi-game. This central role is illustrated by virtually all of the specific results summarized below.

The first result is that equilibrium strategies are always such that the initial allocation appears to be Pareto-efficient. In the economy defined by an equilibrium list of strategies, there is no gain from trade. From this result the important conclusion follows that any equilibrium allocation of the value manipulation quasi-game is also an equilibrium allocation of the (similarly defined) Walrasian manipulation quasi-game. This is a striking result since under truthful behavior, value allocations have in general nothing to do with Walrasian allocations. Also, the true (constrained) Walrasian allocations are always equilibrium allocations. Since the value coincides with the Nash solution in the two-person case, this result is in agreement with the results of Sobel's quoted above. There are, however, other equilibrium allocations; in fact, most equilibrium allocations are not efficient, and the rest of the paper is devoted to their further study.

In the two-person case, a complete characterization is available. Then, the set of equilibrium allocations of the value manipulation quasi-game coincides with the set of equilibrium allocations of the Walrasian manipulation quasi-game. The latter is known from the work of Hurwicz and Otani and Sicilian (the results quoted earlier). For more than two persons, we have the following results: first recall that, from one of our previous conclusions, the net trades corresponding to a given equilibrium allocation belong to a hyperplane. Designating by p the normal to this hyperplane, we show that if the hyperplane is spanned by the convex cone generated by the net trades of some group of agents, then everyone else's consumption is a (constrained) Walrasian consumption relative to p . In an economy in which the number of agents is large as compared to the number of commodities (an example being that of an economy obtained

⁵ We use the term "quasi-game", instead of the more usual term "game", because we have an outcome "correspondence" and not an outcome function. More on this in Section 2.

by multiple replications of a model economy), an equilibrium allocation for which this spanning property holds is therefore *approximately* a (constrained) Walrasian allocation, since the number of agents in a minimal spanning group is bounded by twice the dimensionality of the commodity space.

Finally, we show that two simple respecifications of the value manipulation quasi-game yield *exactly* the (constrained) Walrasian allocations at equilibrium. In these circumstances, not only is efficiency not destroyed by manipulative behavior, but the allocations that result can be precisely identified. The fact that efficiency is preserved significantly mitigates the negative conclusion that the distributional objectives embodied in the value will not be attained.

The paper is organized as follows: Section 2 is devoted to preliminaries – describing the class of economies under consideration, defining the value, setting up the value manipulation quasi-game; Section 3 analyzes unilateral manipulation, and Section 4 joint manipulation; Section 5 is devoted to two reformulations of the value quasi-game, and Section 6 contains some concluding comments.

2 Preliminaries

2.1 Transferable Utility Economies

We consider pure-exchange transferable utility economies with l commodities and n agents. The set of agents is denoted I , with generic element i . A *consumption* z_i for agent i is a pair (x_i, y_i) , with $x_i \in \mathbb{R}$ and $y_i \in \mathbb{R}^{l-1}$. Agent i is characterized by an *initial endowment* $\omega_i \in \mathbb{R}_{++}^l$ and a *utility function* $u_i: Z_i \equiv \mathbb{R} \times \mathbb{R}_+^{l-1} \rightarrow \mathbb{R}$ such that $u_i(z_i) = x_i + v_i(y_i)$ for some continuous, concave and strictly increasing function $v_i: \mathbb{R}_+^{l-1} \rightarrow \mathbb{R}$ satisfying $v_i(\omega_{iy}) = 0$ (this is a normalization condition of no consequence). The class of all such u_i is denoted U_i . The additive separability, and linearity in x_i , of u_i , permits “utility transfers” among the agents. The geometric implication of this special form of the utility functions is that the whole preference map of each individual can be generated from a sample indifference hypersurface by translating it parallel to the x -axis. Frequent use of this fact will be made in our illustrations.

We assume endowments to be known. Therefore, an economy is completely specified by a list $u \equiv (u_1, \dots, u_n) \in U \equiv U_1 \times \dots \times U_n$. The list u will also be written as (u_i, u_{-i}) , where $u_{-i} \equiv (u_1, \dots, u_{i-1}, u_{i+1}, \dots, u_n)$. A *feasible allocation* is a list $z = (z_1, \dots, z_n) \in Z \equiv Z_1 \times \dots \times Z_n$ such that $\sum z_i \leq \sum \omega_i \equiv \Omega$.⁶ The set of feasible

⁶ The symbol $\sum$ indicates a summation over all $i \in S$.

allocations is denoted F . Given $z \in F$, z is *individually rational* for u if $u_i(z_i) \geq u_i(\omega_i)$ for all i ; z is *Pareto-efficient* for u if there does not exist $z' \in F$ with $u_i(z'_i) \geq u_i(z_i)$ for all i , strict inequality holding for at least one i . The sets of individually rational and Pareto-efficient allocations for u are denoted $I(u)$ and $P(u)$ respectively, and their intersection $IP(u)$. Since each u_i is strictly increasing, any interior allocation belonging to $P(u)$ is supported by, and only by, *positive* prices.

2.2 Value Allocations

The *value* is a game-theoretic solution concept introduced by Shapley (1953). Its application to economics appears in Shapley and Shubik (1969). It involves the following definitions:

A *coalition* S is a non-empty subset of I . The *worth* of S , $w^S(u)$, is the maximal aggregate utility that S can achieve through redistribution among its members of the resources at its disposal, namely $\sum_S \omega_i$:

$$\begin{aligned} w^S(u) &\equiv \max \left\{ \sum_{i \in S} u_i(z_i) \mid \sum_{i \in S} z_i \leq \sum_{i \in S} \omega_i \right\} \\ &= \sum_{i \in S} \omega_{ix} + \max \left\{ \sum_{i \in S} v_i(y_i) \mid \sum_{i \in S} y_i \leq \sum_{i \in S} \omega_{iy} \right\}. \end{aligned}$$

Let $Y^S(u)$ be the set of solutions to this last maximization exercise. It is a subset of $\mathbb{R}_+^{(l-1)|S|}$, where $|S|$ denotes the cardinality of S . Also let S_i be the class of coalitions to which agent i belongs. Given $S \in S_i$, the *contribution of agent i to S* is the difference between the worths of the coalitions S and $S \setminus \{i\}$, where the latter is written as $S \setminus i$ with a slight abuse of notation. The *value of agent i* , $\phi_i(u)$, is a weighted average of his contributions to the various elements of S_i , where the weights are combinatorial coefficients adding up to one, specifically:

$$\phi_i(u) = \sum_{S \in S_i} k^S (w^S(u) - w^{S \setminus i}(u)), \quad \text{with} \quad k^S = \frac{(|S| - 1)!(n - |S|)!}{n!}.$$

Finally, a *value allocation* is an element of F at which the utility of each agent is equal to his value. The set of value allocations is denoted $V(u)$:

$$z \in V(u) \Leftrightarrow z \in F, \quad \text{and for all } i, \quad u_i(z_i) = \phi_i(u).$$

2.3 The Value Quasi-game

In order to determine the value allocations of an economy, it is necessary to know the agents' utility functions. However, by announcing a false utility function, an agent may be able to make appear as value allocations, allocations that he prefers. The situation that results when all agents behave strategically, each of them taking into account the distortions due to the others, can be analyzed as a game: the *strategy spaces* are the spaces U_i of admissible utility functions, and the correspondence V , which associates with each list of strategies $u \in U$ the set of value allocations of u , is the *outcome correspondence*. Because of the possible multivaluedness of V , we will in fact refer to the pair (U, V) as the *value "quasi-game"*.

If for some $u \in U$, $V(u)$ is not a singleton, any two allocations z and z' in $V(u)$ are equivalent to all agents according to u , that is, for all i , $u_i(z_i) = u_i(z'_i)$; however, evaluated according to the true utilities $u^0 = (u_1^0, \dots, u_n^0)$, z and z' may not be equivalent. Therefore, multiplicities in $V(u)$ could make it difficult for agents to evaluate their strategic opportunities. Indeed, given an allocation $z \in V(u)$, a strategy u'_i may be available to some agent i such that $V(u'_i, u_{-i})$ contains two allocations z' and z'' with $u_i^0(z'_i) > u_i^0(z_i) > u_i^0(z''_i)$. Assuming that the other agents do not depart from their initial announcements u_{-i} , in the spirit of the usual concept of a Nash-(non-cooperative) equilibrium, which of the two strategies u_i and u'_i will agent i prefer? An answer to this question seems in general to require that agent i 's expectations concerning resolutions of multiplicities be specified. An optimistic agent would expect z' to be selected over z'' , and a pessimistic agent the opposite. Without precisely specifying these expectations, it may not be possible to reach any definite conclusions about the equilibria of the value quasi-game. As it turns out, specifying expectations will not be necessary here, for the following reason: whenever for agent i some strategy u'_i exists as described above, then some other strategy u''_i is also available to him with $u_i^0(z''_i) > u_i^0(z)$ for all $z'' \in V(u''_i, u_{-i})$. This observation motivates our choice of equilibrium concept; it is, in our context, the natural counterpart of the usual concept of a Nash equilibrium.

Definition: A pair $(u, z) \in U \times Z$ with $z \in V(u)$ is an *equilibrium of the quasi-game* (U, V) for u^0 if

$$\forall i \in I, \quad \forall u'_i \in U_i, \quad \forall z' \in V(u'_i, u_{-i}), \quad u_i^0(z_i) \geq u_i^0(z'_i).$$

The set of equilibria of (U, V) for u^0 is denoted $E(U, V; u^0)$. If (u, z) is an equilibrium, u is called an *equilibrium list of strategies* and z an *equilibrium allocation*.

Additional notation: $T(u_i, z_i) \equiv \{z'_i \in Z_i | u_i(z'_i) \geq u_i(z_i)\}$. $T^+(u_i, z_i) \equiv \{z'_i \in Z_i | u_i(z'_i) > u_i(z_i)\}$. $\Delta^{l-1} \equiv \{p \in \mathbb{R}_+^l | \sum p_i = 1\}$. $H(u_i, z_i) \equiv \{p \in \Delta^{l-1} | \forall z'_i \in T(u_i, z_i), pz'_i \geq pz_i\}$. Given $p \in \Delta^{l-1}$, $B_i(p) \equiv \{z_i \in Z_i | pz_i \leq p\omega_i\}$, and $d_i(u_i, p) \equiv \{z_i \in Z_i | \forall z'_i \in B_i(p), u_i(z_i) \geq u_i(z'_i)\}$. Given u_{-i} , $h_i(u_{-i}) \equiv \{z_i \in Z_i | \exists p \in \Delta^{l-1} \text{ and for each } j \neq i, z_j \in d_j(u_j, p) \text{ such that } z_i - \omega_i = - \sum_{j \neq i} (z_j - \omega_j)\}$.

After introducing allocations such as z or $\tilde{z}$, we will often use symbols such as x_i or $\tilde{y}_i$ without further explanation. z and $\tilde{z}$ should then respectively be understood as $z = (z_1, \dots, z_n)$, and $\tilde{z} = (\tilde{z}_1, \dots, \tilde{z}_n)$, or more precisely $((x_1, y_1), \dots, (x_n, y_n))$, and $((\tilde{x}_1, \tilde{y}_1), \dots, (\tilde{x}_n, \tilde{y}_n))$.

3 Unilateral Manipulation

The purpose of this section is two-fold; first, to give a geometric description of the case $(l, n) = (2, 2)$. When $n = 2$, the value is easy to compute (it coincides with the Nash bargaining solution), and we will find this simple case useful to illustrate several of our results. Second, to provide an analysis of unilateral manipulation of the value parallel to the standard analysis of unilateral manipulation of the price mechanism. Unambiguous conclusions about the relative manipulability of the two mechanisms will be obtained. In the process, we will uncover the essential intuition behind the most important result of the next section, which deals with joint manipulation, namely, that in general optimal strategies are linear functions.

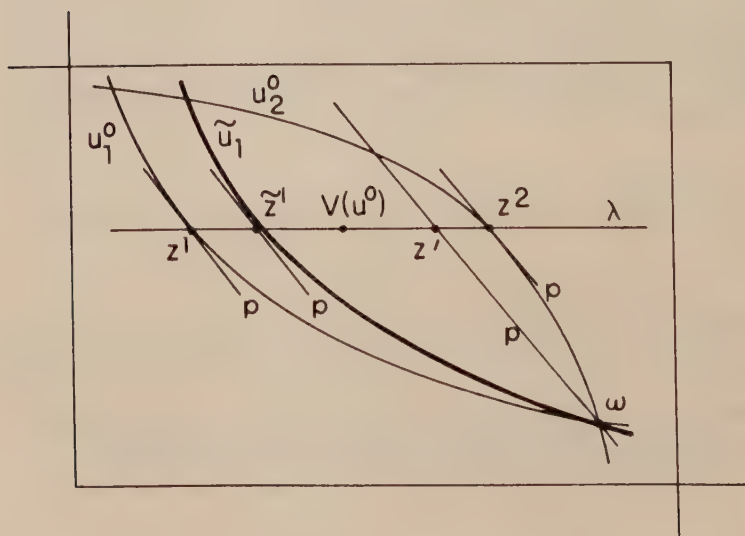


Fig. 1

Assume then that $(l, n) = (2, 2)$. In Fig. 1, we have represented the agents' true indifference curves through ω — they are labeled u_1^0 and u_2^0 — and the Pareto-efficient set of the true economy $P(u^0)$. The transferable utility property implies that $P(u^0)$ is a horizontal line (if the maximizer of $v_1^0(y_1) + v_2^0(y_2)$ subject to $y_1 + y_2 \leq \omega_{1,y} + \omega_{2,y}$ is unique) or a horizontal band (if this maximizer is not unique). In the example of Fig. 1, the indifference curves through ω are drawn strictly convex (that is, v_1^0 and v_2^0 are assumed to be strictly concave), so that $P(u^0)$ is a horizontal line, labeled λ . The efficient allocation that is indifferent to ω for agent 1 according to u_1^0 is denoted z^1 ; z^2 is defined analogously. The utility gains that the agents can achieve by trading from ω can be meaningfully measured in terms of the x -commodity by the horizontal distance between z^1 and z^2 . There is a unique value allocation, which is the midpoint of the segment $[z^1, z^2]$: $V(u^0) = \{(z^1 + z^2)/2\}$. This is the allocation that would be chosen if both agents told the truth.

We now assume that only agent 2 tells the truth and we determine *which allocations agent 1 can make appear as value allocations by announcing some admissible utility u_1* . We will say that these allocations are *achievable* by agent 1. (If an allocation is achievable as *unique* value allocation, we will say that it is *enforceable* by agent 1.) For instance, what points of λ are achievable by agent 1?

Let $p \in H(u_2^0, z_2^2)$ and $\tilde{z}^1$ be a point of λ slightly to the right of z^1 . Also let $\tilde{u}_1 \in U_1$ be such that (i) $\tilde{u}_1(\tilde{z}^1) = \tilde{u}_1(\omega_1)$, (ii) $p \in H(\tilde{u}_1, \tilde{z}^1)$, and (iii) $\tilde{v}_1$ be strictly concave at y_1 . It follows from (ii) and (iii) that $P(\tilde{u}_1, u_2^0) = \lambda$ and that $V(\tilde{u}_1, u_2^0)$ is a singleton, the midpoint of $[\tilde{z}^1, z^2]$; since this point is slightly to the right of $V(u^0)$, agent 1 has gained by cheating. How far can he so push the value allocation to the right of $V(u^0)$ on λ ?

To answer this question, first identify $p \in H(u_2^0, z_2^2)$ such that the point z' of λ defined by $pz_2' = p\omega_2$ assumes its rightmost position on λ . Note that $z_2' \in d_2(u_2^0, p)$. The rightmost position of $\tilde{z}^1$ that permits the selection of $\tilde{u}_1$ satisfying requirements (i) and (ii) of the paragraph above is precisely z' , and the resulting value allocation is the midpoint of $[z', z^2]$.

If v_2^0 were not strictly concave at y_2^2 , the midpoint of $[z', z^2]$ would not be the only element of $V(\tilde{u}_1, u_2^0)$ since then, requirement (iii) could not be imposed on $\tilde{u}_1$. But the midpoint of $[z', z^2]$ could be arbitrarily closely approached by a sequence of enforceable value allocations (corresponding to a sequence of choices $\{z^{1v}\}$ converging to z' from the left).

Although we just focused on points of $\lambda = P(u^0)$, the argument would be exactly the same for any horizontal line, yielding the following characterization of the rightmost boundary of the set of points achievable by agent 1: This is simply the set of points that are "halfway" (horizontally) between agent 2's true indifference curve through ω and his true offer curve $h_1(u_2^0)$. (This set is the dashed line of Fig. 2.)

Unilateral and optimal manipulation of the value by agent 1 involves maximizing u_1^0 on this locus. Because the locus is by construction always to the right of $h_1(u_2^0)$, and because $h_1(u_2^0)$ is the locus of allocations that agent 1 can achieve under manipula-

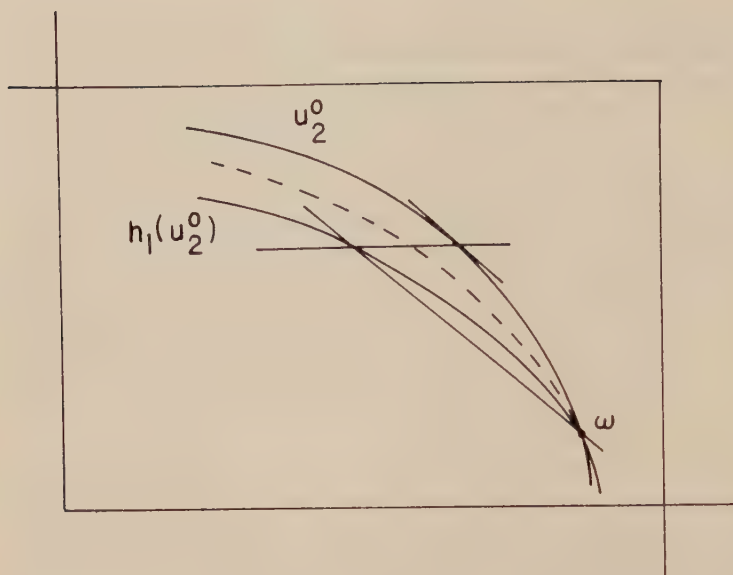


Fig. 2

tion of the Walrasian mechanism, we obtain the following theorem, which is easily shown to hold for arbitrary (l, n) .

Theorem 1: Unilateral and optimal manipulation of the value is always more profitable than unilateral and optimal manipulation of the Walrasian mechanism.

Does Theorem 1 imply that the incentive to manipulate the value is greater than the incentive to manipulate the Walrasian mechanism? Not necessarily, since there are other ways to evaluate the benefits of manipulation. Indeed, the other agents could be cheating too, a possibility that motivates our next section on joint manipulation. In addition, the utility *gain* that an agent can achieve by unilateral and optimal manipulation as compared to behaving honestly, may be a more useful criterion. Note that it is a meaningful one, since this gain can be measured in terms of the x -commodity, again thanks to the transferable utility assumption, and that comparisons of mechanisms on the basis of it are legitimate for the same reason. However, the criterion does not allow an unambiguous ranking of the value and of the Walrasian mechanism.

Theorem 2: The gain (measured in terms of the x -commodity) obtained from unilateral and optimal manipulation of the value may be greater or smaller than the gain (similarly measured) obtained from unilateral and optimal manipulation of the Walrasian mechanism.

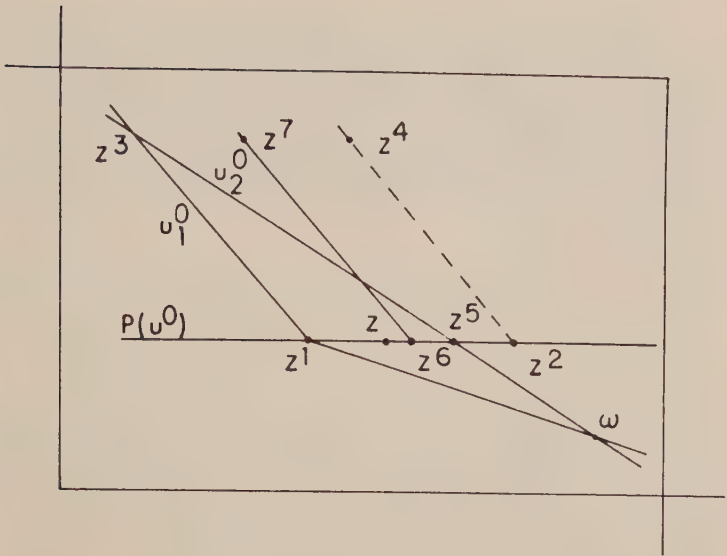


Fig. 3

Proof: The proof is by way of an example with $(l, n) = (2, 2)$, represented in Fig. 3. Agent 1's true indifference curve through ω is the broken line $\omega z^1 z^3$ (labeled u_1^0), and the corresponding offer curve is the broken line $\omega z^1 z^2 z^4$. Agent 2's true indifference curve through ω is the straight line ωz^3 (labeled u_2^0), and this is also his derived offer curve. The Pareto set for the true economy is the horizontal line through z^1 (labeled $P(u^0)$). The true economy has a unique value allocation $z \equiv (z^1 + z^5)/2$ and a unique Walrasian allocation z^5 . Agent 1 can make appear all points of ωz^3 both as value allocations and as Walrasian allocations, and among those, the point that he prefers is z^5 . Therefore, the gain that he can achieve through unilateral and optimal manipulation of the value is $x_1^5 - x_1 > 0$, while the corresponding gain for the Walrasian mechanism is zero. Similarly, agent 2 can make appear as value allocation any point of the broken line $\omega z^1 z^6 z^7$, and as Walrasian allocation any point of the broken line $\omega z^1 z^2 z^4$. This implies that the gain that he can achieve through unilateral and optimal manipulation of the value is $x_2^1 - x_2$, while the corresponding gain for the Walrasian mechanism is $x_2^1 - x_2^5 = 2(x_2^1 - x_2)$. Q.E.D.

4 Characterizing the Equilibrium Allocations of the Value Manipulation Quasi-game

We are now ready to undertake the study of the equilibria of the value quasi-game. In order to simplify the presentation, we limit our attention to interior allocations, some remarks on boundary allocations being left to the concluding section.

First, we establish an essential property of equilibrium lists of strategies: *they are such that the initial allocation appears to be Pareto-efficient* (Lemma 1). We also show that *any equilibrium allocation of the value quasi-game is an equilibrium allocation of the (analogously defined) Walrasian quasi-game* (Proposition 2), and that *the true Walrasian allocations are equilibrium allocations of the value quasi-game* (Proposition 3). The first lemma says that, at equilibrium, there is no gain from trade in the apparent economy.

Lemma 1: If (u, z) is an equilibrium of (U, V) for u^0 , then ω is Pareto-efficient for u .

Proof: Let $(u, z) \in U \times Z$ with $z \in V(u)$ be given. Since $V(u) \subset P(u)$, there exists $p \in S^{l-1}$ supporting z in u . For each i , let $z_i^1 \in Z_i$ be such that $u_i(z_i^1) = u_i(\omega_i)$ and $y_i^1 = y_i$. Since z_i and z_i^1 differ only in their first coordinates, the quasi-linear form of the utilities implies that $p \in H(u_i, z_i^1)$. By convexity of preferences, $p z_i^1 \leq p \omega_i$. We want to show that if $(u, z) \in E(U, V; u^0)$, then $p z_i^1 = p \omega_i$ for each i . We assume, by way of contradiction, that for one agent, say agent 1, the inequality is strict: $p z_1^1 < p \omega_1$, and we establish the existence of a strategy $\tilde{u}_1 \in U_1$ that is a better strategy for him than u_1 . The argument is illustrated in Fig. 4.

Let $\tilde{z}_1^1 \equiv (x_1^1 + \epsilon, y_1)$ with ϵ positive but small enough to guarantee that $p \tilde{z}_1^1 < p \omega_1$, and let $\tilde{u}_1 \in U_1$ be such that

$$(i) \quad \tilde{u}_1(\tilde{z}_1^1) = \tilde{u}_1(\omega_1),$$

$$(ii) \quad p \in H(\tilde{u}_1, \tilde{z}_1^1),$$

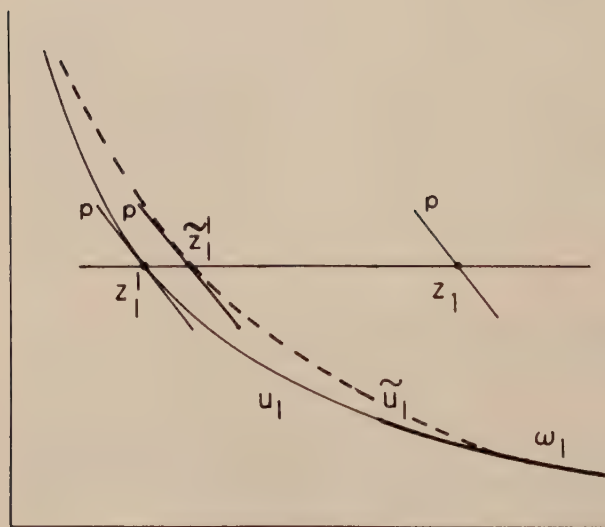


Fig. 4

(iii) $\tilde{v}_1$ is strictly concave at y_1 , and

(iv) $v_1 - \tilde{v}_1 \leq \epsilon$.

It follows from (ii) and (iii) that, for all $\tilde{z} \in P(\tilde{u}_1, u_{-1})$, $\tilde{y}_1 = y_1$. Also, from (i), it follows that equality holds in (iv) for the value y_1 of the argument.

Let $S \in S_1$, $y^S \in Y^S(u)$ and $\tilde{y}^S \in Y^S(\tilde{u}_1, u_{-1})$ be given. First, we observe that by definition of $w^S(\tilde{u}_1, u_{-1})$:

$$\begin{aligned} \sum_{i \in S} \omega_{ix} + \tilde{v}_1(y_1^S) + \sum_{\substack{i \in S \\ i \neq 1}} v_i(y_i^S) &\leq \sum_{i \in S} \omega_{ix} + \tilde{v}_1(\tilde{y}_1^S) \\ &+ \sum_{\substack{i \in S \\ i \neq 1}} v_i(\tilde{y}_i^S) = w^S(\tilde{u}_1, u_{-1}). \end{aligned}$$

This implies that

$$\begin{aligned} w^S(u) - w^S(\tilde{u}_1, u_{-1}) &\leq \sum_{i \in S} v_i(y_i^S) - [\tilde{v}_1(y_1^S) + \sum_{\substack{i \in S \\ i \neq 1}} v_i(y_i^S)] \\ &= v_1(y_1^S) - \tilde{v}_1(y_1^S) \leq \epsilon. \end{aligned}$$

the last inequality following from (iv).

Also, $w^{\{1\}}(u) - w^{\{1\}}(\tilde{u}_1, u_{-1}) = u_1(\omega_1) - \tilde{u}_1(\omega_1) = 0$.

Finally, we observe that $w^{S \setminus 1}(u) - w^{S \setminus 1}(\tilde{u}_1, u_{-1}) = 0$ for all S . We can now compute the change in agent 1's value as

$$\begin{aligned} \phi_1(u) - \phi_1(\tilde{u}_1, u_{-1}) &= \sum_{S \in S_1} k^S [w^S(u) - w^{S \setminus 1}(u)] - \sum_{S \in S_1} k^S [w^S(\tilde{u}_1, u_{-1}) \\ &- w^{S \setminus 1}(\tilde{u}_1, u_{-1})] = k^{\{1\}} (w^{\{1\}}(u) - w^{\{1\}}(\tilde{u}_1, u_{-1})) \\ &+ \sum_{\substack{S \in S_1 \\ S \neq \{1\}}} k^S [w^S(u) - w^S(\tilde{u}_1, u_{-1}) + w^{S \setminus 1}(u) - w^{S \setminus 1}(\tilde{u}_1, u_{-1})] \\ &= \sum_{\substack{S \in S_1 \\ S \neq \{1\}}} k^S [w^S(u) - w^S(\tilde{u}_1, u_{-1})] \leq \sum_{\substack{S \in S_1 \\ S \neq \{1\}}} k^S \epsilon < \epsilon, \end{aligned}$$

where the last inequality follows from the facts that $k^{\{1\}} > 0$ and $\sum_{S \in S_1} k^S = 1$.

From (ii) and (iii) it follows that if $\tilde{z} \in P(\tilde{u}_1, u_{-1})$, then $\tilde{y}_1 = \bar{y}_1 (= y_1)$. Since $V \subset P$, the same conclusion holds for any $\tilde{z} \in V(\tilde{u}_1, u_{-1})$. Finally, since $\tilde{v}_1(y_1) = v_1(y_1) - \epsilon$, we have that

$$\tilde{u}_1(z_1) = u_1(z_1) - \epsilon = \phi_1(u) - \epsilon < \phi_1(\tilde{u}_1, u_{-1}).$$

Therefore a consumption of the form $(\tilde{x}_1, y_1)$ achieves agent 1's new value only if $\tilde{x}_1 > x_1$, which implies that $u_1^0(\tilde{x}_1, y_1) > u_1^0(z_1)$, and that (u, z) is not an equilibrium of (U, V) for u^0 . Q.E.D.

The next Proposition states a property of equilibrium lists of strategies in terms of the true utilities. The Walrasian correspondence starts playing a crucial role here.

Proposition 1: At an equilibrium (u, z) of (U, V) for u^0 , the net offer surface faced by each agent i , determined according to the announced utilities u_{-i} of the other agents, does not contain any consumption that he truly prefers to z_i .

Proof: Let $(u, z) \in U \times Z$ with $z \in V(u)$ be given and suppose that

$$\exists i, \quad \exists p \in \Delta^{l-1}, \quad \text{and} \quad \forall j \neq i, \quad \exists z'_j \in d_j(u_j, p) \quad \text{s.t.}$$

$$u_i^0(z'_i) > u_i^0(z_i) \quad \text{for} \quad z'_i \equiv \Omega - \sum_{j \neq i} z'_j.$$

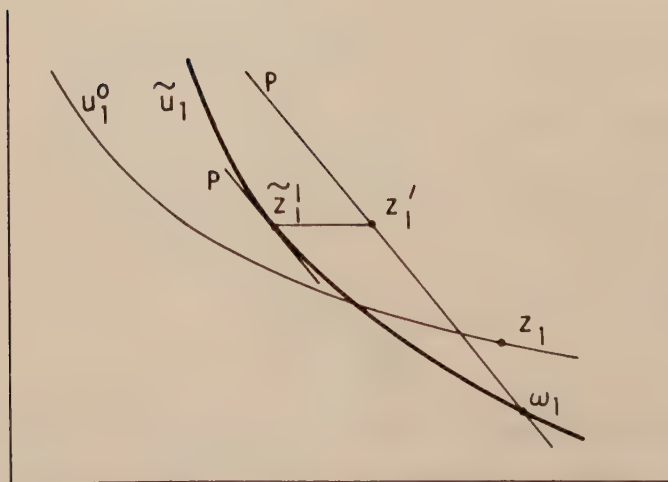


Fig. 5

The proof that $(u, z) \notin E(U, V; u^0)$ is illustrated in Fig. 5.

Suppose, without loss of generality, that $i = 1$. Then, for ϵ positive but small enough, $u_1^0(\tilde{z}_1^1) > u_1^0(z_1)$ where $\tilde{z}_1^1 \equiv (x_1' - \epsilon, y_1')$. Now, let $\tilde{u}_1 \in U_1$ be such that

$$(i) \tilde{u}_1(\tilde{z}_1^1) = \tilde{u}_1(\omega_1),$$

$$(ii) p \in H(\tilde{u}_1, \tilde{z}_1^1), \quad \text{and}$$

$$(iii) \tilde{v}_1 \text{ is strictly concave in a neighborhood of } \tilde{y}_1^1 = y_1'.$$

It follows from (ii), (iii), and the fact that $z_1' \in h_1(u_{-1})$, that if $\tilde{z} \in P(\tilde{u}_1, u_{-1})$, then $\tilde{y}_1 = y_1'$. This fact and (i) imply that if $\tilde{z} \in IP(\tilde{u}_1, u_{-1})$ then $\tilde{y}_1 = y_1'$ and $\tilde{x}_1 \geq \tilde{x}_1^1$. Therefore, since V is a subcorrespondence of IP , it follows that if $\tilde{z} \in V(\tilde{u}_1, u_{-1})$, then $u_1^0(\tilde{z}_1) \geq u_1^0(\tilde{z}_1^1) > u_1^0(z_1)$, and consequently $(u, z) \notin E(U, V; u^0)$. Q.E.D.

Let (U, W) be the manipulation quasi-game derived from the Walrasian correspondence W as (U, V) was derived from the value correspondence. The next Proposition relates the equilibrium allocations of the two quasi-games.

Proposition 2: Any equilibrium allocation of (U, V) for u^0 is an equilibrium allocation of (U, W) for u^0 .

Proof: From Lemma 1, it follows that if $(u, z) \in E(U, V; u^0)$, then $z \in W(u)$ or equivalently, for each i , $z_i \in h_i(u_{-i})$. Then, Proposition 1 says that for each i , z_i maximizes u_i^0 on $h_i(u_{-i})$. In addition, under the operation of W , only points of $h_i(u_{-i})$ are attainable by agent i through the announcement of some $u_i \in U_i$. Therefore $(u, z) \in E(U, W; u^0)$. Q.E.D.

Proposition 3: The Walrasian allocations of u^0 are equilibrium allocations of (U, V) for u^0 .

Proof: Let $z \in W(u^0)$, with $p \in \Delta^{I-1}$ as associated Walrasian price, be given. For each i , let $u_i \in U_i$ be such that (i) $u_i(z_i) = u_i(\omega_i)$ and (ii) $p \in H(u_i, z_i)$. We claim that $(u, z) \in E(U, V; u^0)$. We prove the claim by showing that agent 1 — this is without loss of generality — cannot attain an allocation that he truly prefers to z , through a unilateral change in his strategy. Indeed, for all $\tilde{u}_1 \in U_1$, if $\tilde{z} \in V(\tilde{u}_1, u_{-1})$, then $\tilde{z} \in I(\tilde{u}_1, u_{-1})$ and this implies that for all $i \neq 1$, $p\tilde{z}_i \geq p\omega_i$, and in turn $p\tilde{z}_1 \leq p\omega_1$, but since $z \in W(u^0)$, this yields $u_1^0(z_1) \geq u_1^0(\tilde{z}_1)$. Q.E.D.

An analogue for Proposition 3 for a class of 2-person bargaining solutions containing the Nash (1950) and the Kalai-Smorodinsky (1975) solutions is proved by Sobel

(1981). Sobel, however, does not restrict his attention to transferable utility economies. Proposition 3 is somewhat encouraging since it says that manipulation does not necessarily lead to a violation of efficiency. Unfortunately, there are in general equilibrium allocations that are not efficient; it is the object of the next pages to pursue their characterization further.

Lemma 2: At an equilibrium (u, z) of (U, V) for u^0 , for no agent i does there exist $\{\lambda_j\}_{j \neq i} \in [0, 1]^{n-1}$ such that $\omega_i - \sum_{j \neq i} \lambda_j(z_j - \omega_j) \in T^+(u_i^0, z_i)$. In particular $[\omega_i, z_i] \cap T^+(u_i^0, z_i) = \emptyset$.

Proof: Let $(u, z) \in E(U, V; u^0)$ be given. From Lemma 1, it follows directly that for some $p \in \Delta^{l-1}$, for each j , and for each $\lambda_j \in [0, 1]$, $\omega_j + \lambda_j(z_j - \omega_j) \in d_j(u_j, p)$. This implies that for all i , and for all $\{\lambda_j\}_{j \neq i} \in [0, 1]^{n-1}$, $\omega_i - \sum_{j \neq i} \lambda_j(z_j - \omega_j) \in h_i(u_i)$.

The proof of the first claim concludes by an application of Proposition 1. The second claim follows from the first by taking all the λ_j equal. Q.E.D.

Lemma 2 is illustrated in Fig. 6 for $(l, n) = (3, 3)$. The pair (u, z) is assumed to be an equilibrium of (U, V) for u^0 . The net trades $t_i \equiv z_i - \omega_i$, for $i = 1, 2, 3$, lie in a hyperplane H through 0 normal to p , a price of support to $z \in P(u)$. The shaded area A represents all the vectors of the form $\lambda_2(z_2 - \omega_2) + \lambda_3(z_3 - \omega_3)$ where $\lambda_2, \lambda_3 \in [0, 1]$.

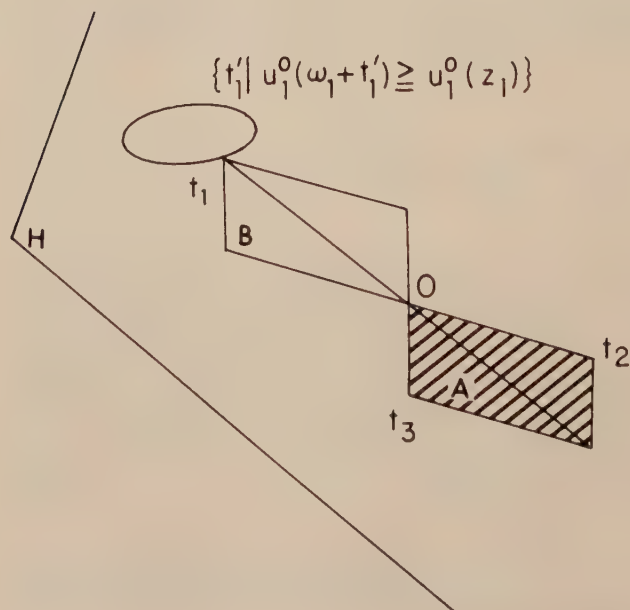


Fig. 6

1]. Any such vector is a net trade vector that agent 2 and agent 3 would collectively desire at prices p according to their announced utilities. Area B is the symmetric image of A with respect to the origin. Lemma 2 says that this area should not contain any net trade that agent 1 truly prefers to t_1 . This is indicated in the figure by drawing the intersection with H of the set of net trades that agent 1 truly prefers to t_1 as bounded by a curve that does not intersect B .

Lemma 2 permits the identification, in terms of the true preferences u^0 , of a subset $L(u^0)$ of the feasible set that always contains the set of equilibrium allocations of the value quasi-game.

$$L(u^0) = \{z \in F \mid \text{(i) } \exists p \in \Delta^{l-1} \text{ s.t. } pz_i = p\omega_i \forall i, \text{ (ii) } \forall i, z_i \in [\omega_i, z_i^*], \\ \text{where } z_i^* \text{ is a maximizer of } u_i^0 \text{ on the set of consumptions } z'_i \in Z_i \\ \text{of the form } \omega_i + \mu(z_i - \omega_i) \text{ for } \mu \geq 0\}.$$

If $(l, n) = (2, 2)$, $L(u^0)$ is the area (in the feasible set) delimited by the true offer curves. If $n = 2$ but l is arbitrary, $L(u^0)$ is the "hyper-volume" (in the feasible set), delimited by the true offer "hyper-surfaces".

Proposition 4: The set of equilibrium allocations of (U, V) for u^0 is a subset of $L(u^0)$.

Proof: Let $(u, z) \in E(U, V; u^0)$ be given. First, we note that (i) of the definition of $L(u^0)$ holds, by Lemma 1. To prove (ii), note that if this condition did not hold, there would be some agent i and some $z'_i \in [\omega_i, z_i]$ that agent i would strictly prefer to z_i according to his true preferences, in contradiction with the second statement of Lemma 2. Q.E.D.

If $n = 2$, a complete characterization of the equilibrium allocations of (U, V) can be obtained.

Proposition 5: Let $n = 2$. Then the set of equilibrium allocations of (U, V) for u^0 coincides with $L(u^0)$.

Proof: In view of proposition 4, it suffices to show that if $z \in L(u^0)$, there exists $u \in U$ such that $(u, z) \in E(U, V; u^0)$. So, let $z \in L(u^0)$, $p_1 \in H(u_1^0, z_1)$ and $p_2 \in H(u_2^0, z_2)$ be given. By definition of $L(u^0)$, we have that $p_1 z_1 \geq p_1 \omega_1$ and $p_2 z_2 \geq p_2 \omega_2$. Let now $p \in \Delta^{l-1}$ be such that $p z_1 = p \omega_1$ (and therefore $p z_2 = p \omega_2$), and for all i , let $u_i \in U_i$ be such that (i) $u_i(z_i) = u_i(\omega_i)$ and (ii) $p, p_j (j \neq i) \in H(u_i, z_i)$. It is immediate that $z \in V(u) = IP(u)$. Also, for all $\tilde{z} \in F$, if $u_i(\tilde{z}_i) \geq u_i(z_i)$, then $u_j^0(\tilde{z}_j) \leq$

$u_j^0(z_j)$. This means that for all $\tilde{u}_j \in U_j$, if $\tilde{z} \in V(u_i, \tilde{u}_j)$, which implies that $\tilde{z} \in I(u_i, \tilde{u}_j)$, then $u_j^0(\tilde{z}_j) \leq u_j^0(z_j)$, and agent j cannot gain by changing his strategy from u_j to $\tilde{u}_j$. Q.E.D.

This result should be compared to the one obtained by Hurwicz (1972) in his study of the manipulability of the Walrasian mechanism when $(l, n) = (2, 2)$, and to the generalization of his conclusions to arbitrary l by Otani and Sicilian (1982): in Hurwicz's study, strategies are offer curves consistent with (that is, derivable from) smooth (C^2) and strictly convex preferences, and only strategy lists yielding a unique Walrasian allocation are considered; the set of Nash equilibrium allocations of this "generalized" game coincides with the interior of the area delimited by the true offer curves. This is precisely $\text{int. } L(u^0)$. (It can be shown that the boundary of $L(u^0)$ too belongs to the equilibrium set if the smoothness requirement is removed; furthermore, a similar characterization obtains if multiplicities of Walrasian allocations are permitted but are dealt with as in the present paper.) The technical conditions used by Otani and Sicilian are somewhat different but these authors too find the set of equilibrium allocations of the Walrasian quasi-game that they formulate to be $L(u^0)$. Note that equilibrium allocations are individually-rational for the true preferences. Individual rationality is indeed guaranteed by announcing one's true utility, which is one of the permissible choices.

Let us now return to the case of arbitrary n . From Proposition 3, we know that the true Walrasian allocations are always equilibrium allocations. The next proposition gives an upper bound on the number of agents whose consumptions at equilibrium are not Walrasian relative to the price supporting the allocation. This result will imply that if the number of consumers is large as compared to the number of commodities, any equilibrium allocation whose associated net trades span a hyperplane in $\mathbb{R}^l$, will in fact be approximately Walrasian.

Definition: Given a linear subspace Q of $\mathbb{R}^l$, $z_i \in \mathbb{R}^l$ is a Q -Walrasian consumption for i relative to u_i , if z_i maximizes u_i in $\{z'_i \in Z_i | z'_i - \omega_i \in Q\}$. If Q is a hyperplane in $\mathbb{R}^l$, this notion of maximization coincides with the usual maximization of utility in a budget set. This definition is used by Mas-Colell (1982).

Proposition 6: Assume $n > 2l$. Let (u, z) be an equilibrium of (U, V) for u^0 . Let Q be the linear subspace spanned by the convex cone generated by the net trade vectors $\{z_j - \omega_j | j = 1, \dots, n\}$, and let l_Q be the dimensionality of Q . Then, for at least $n - 2l_Q$ agents, z_i is a Q -Walrasian consumption for i relative to u_i^0 .

Proof: If Q is of dimensionality l_Q , it is spanned by the convex cone generated by only $2l_Q$ members of $\{z_j - \omega_j | j = 1, \dots, n\}$. Let S be a set of agents whose trades

have this spanning property. Lemma 2 implies that (u, z) is an equilibrium of (U, V) for u^0 only if for all $i \notin S$, $(\{\omega_i\} + Q) \cap T_i^+(u_i^0, z_i) = \emptyset$. But this says that z_i is Q -Walrasian for i relative to u_i^0 .
Q.E.D.

Remark: The condition $n > 2l$ is imposed only to guarantee that $n - 2l_Q \geq 0$, an inequality that permits us a slightly simpler statement of the Proposition.

Proposition 6 is of particular interest if Q is a hyperplane, that is if the space of "active markets" is of dimension $l - 1$. (We know from Lemma 1 that in fact Q is at most of dimension $l - 1$.) Then, if $p \in \Delta^{l-1}$ is normal to this hyperplane, we can conclude that the consumptions of $n - 2(l - 1)$ agents are Walrasian relative to p . If, in addition, n is large as compared to l , the allocation can be said to be *approximately Walrasian*.

It may of course be the case that for each agent, the linear space spanned by the convex cone generated by the net trades of the other agents has dimensionality $l - 1$. Then, the equilibrium allocation is Walrasian for the true economy. In the particular situation in which the economy is obtained by replication of some model economy, an equilibrium allocation at which identical agents get identical consumptions and the linear space spanned by the convex cone generated by the net trades of *some* group of agents has dimension $l - 1$, is a Walrasian allocation for the true economy.

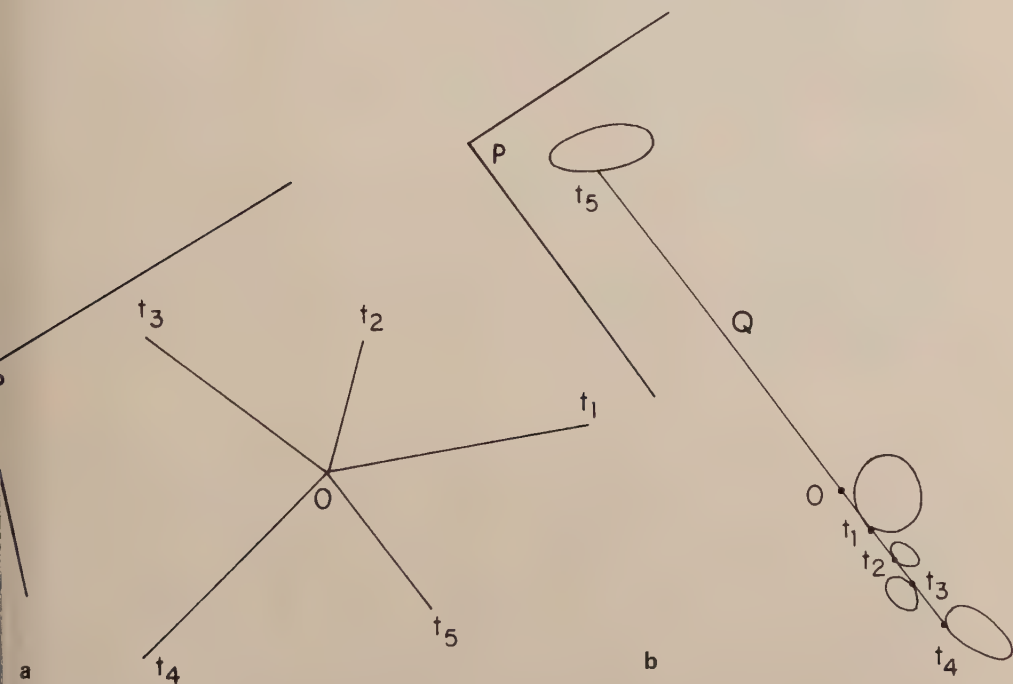


Fig. 7

Proposition 6 and the comments following it are illustrated in the two examples of Fig. 7, where $(l, n) = (3, 5)$, and (u, z) is assumed to be an equilibrium of (U, V) for u^0 . Since $z \in P(u)$, there is $p \in \Delta^{l-1}$ supporting z in u . Let P be the hyperplane normal to p . The net trades $\{z_i - \omega_i | i = 1, \dots, 5\}$ all lie in P . In the first example, illustrated in Fig. 7a, for each i , the convex cone generated by the net trades $\{z_j - \omega_j | j \neq i\}$ happens to span P , and therefore z is a Walrasian allocation for u^0 . In the second example, illustrated in Fig. 7b, there is a one-dimensional space, Q , a subspace of P , of active trading, and by Proposition 6, for at least $n - 2 = 3$ agents, z_i maximizes u_i^0 in $\{\omega_i\} + Q$. This is indicated by the fact that for at least $n - 2$ agents, the set of trades in P that would lead agent i to a higher utility is tangent to Q . As in Fig. 6, only the boundary of each such set is represented. (In fact, for the case illustrated in Fig. 7b, the tangency condition holds for 4 agents; this is because for each of agents 1, 2, 3, 4, the convex cone generated by the net trades of the remaining agents span Q .)

5 Reformulation of the Value Manipulation Quasi-game

It is shown here that under two alternative and minor reformulations of the value manipulation game, *the set of equilibrium allocations reduces to the set of true Walrasian allocations*. This result shows, in particular, that efficiency is not necessarily destroyed by manipulative behavior, although the distributional objectives embodied in the value will obviously not be achieved. The question of the existence and the construction of games whose set of equilibrium allocations always coincides with the set of φ -optimal allocations for an a priori given correspondence φ — such a game being said to *implement* φ — has been extensively studied recently (see in particular Maskin 1977 and Dasgupta, Hammond, and Maskin 1979). The implementation of the Walrasian correspondence was first shown possible by Hurwicz (1979) and Schmeidler (1980). The present paper shows that this correspondence can also be implemented by variants of the value quasi-game.

(a) The first reformulation involves a redefinition of the strategy spaces, strategies being required to satisfy a smoothness condition. This reformulation can be interpreted in two ways. Either the class of economies in which the quasi-game is played is also restricted by the same condition, in which case strategy spaces remain spaces of potential characteristics, and the quasi-game remains a *direct* quasi-game. Or the restriction is not imposed on the true utilities, but then, telling the truth may not be an option to some individuals, and the quasi-game becomes an *abstract* quasi-game. Since, in the statement of Proposition 7, u^0 is still supposed to be an arbitrary element of U , the proposition should be read with the second interpretation in mind.

For each i , let U_i^s be the subset of U_i of utility functions whose derived offer surfaces are smooth, and let $U^s \equiv U_1^s \times \dots \times U_n^s$.

Proposition 7: The set of equilibrium allocations of (U^s, V) for u^0 coincides with the set of Walrasian allocations of u^0 .

Proof: We can prove, as in Lemma 1, that $(u, z) \in U^s \times Z$ with $z \in V(u)$ is an equilibrium of (U^s, V) for u^0 only if $\omega \in P(u)$, which implies that for some p , and for all j , $d_j(u_j, p)$ contains $[0, z_j]$. Then, for all i , (i) $h_i(u_{-i})$ contains $[0, z_i]$. One can also prove, as in Proposition 1, that if $(u, z) \in U^s \times Z$ with $z \in V(u)$ is an equilibrium of (U^s, V) for u^0 , then (ii) for all i , $h_i(u_{-i}) \cap T^+(u_i^0, z_i) = \emptyset$. We now claim that for all i , $z_i \in d_i(u_i^0, p)$, and therefore, that $z \in W(u^0)$. If not, then, for some i , agent i 's true indifference curve through z_i is transversal to his budget plane relative to p . But since $h_i(u_{-i})$ is smooth, (i) implies that there are points of $h_i(u_{-i})$ that belong to $T^+(u_i^0, z)$, in contradiction with (ii). Q.E.D.

Proposition 7 is illustrated in Fig. 8 for $(l, n) = (2, 2)$. The candidate equilibrium (u, z) is such that $z \in V(u)$ for $u \in U$ with $\omega \in P(u)$. Let $p \in \Delta^1$ be the price of support to z . The offer curve $h_1(u_2)$ faced by agent 1 contains the segment $[\omega_1, z_1]$, and by the smoothness assumption, it is tangential to the budget line relative to p at z (as drawn in the figure). Now, if agent 1's true indifference curve through z were

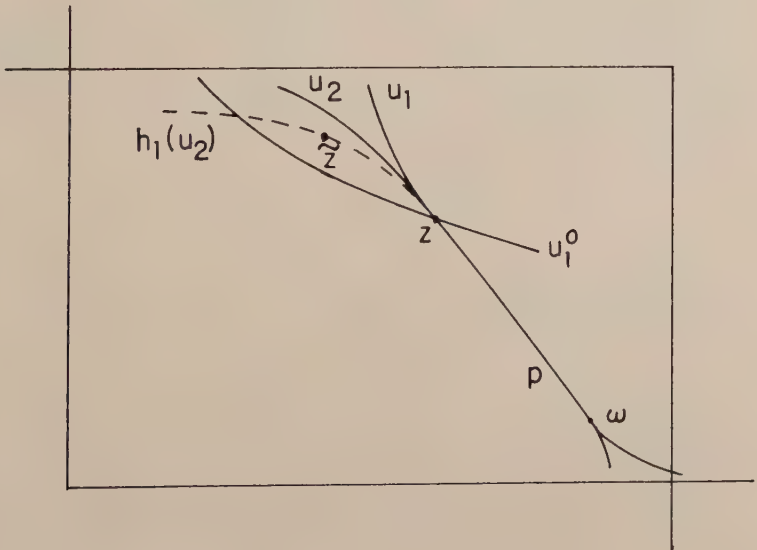


Fig. 8

transversal at z to the budget line, there would be points of $h_1(u_2)$, (such as $\tilde{z}$ in the figure), that agent 1 would strictly prefer to z , according to his true preferences. This would imply that $z \notin E(U^s, V; u^0)$.

(b) The other reformulation of the value manipulation quasi-game involves a redefinition of both the strategy spaces and of the outcome correspondence, and it can be given the same two interpretations as the first one.

For each i , let $\bar{U}_i$ be the subset of U_i of strategies u_i such that $H_i(u_i, \omega_i)$ contains a unique point p^i and the cone spanned by the set $A(u_i) \equiv \{z_i \in Z_i | p^i z_i = p^i \omega_i, u_i(z_i) = u_i(\omega_i)\}$ is a linear subspace of dimensionality $l - 1$. Then, let $A^0(u_i)$ be the interior of $A(u_i)$ relative to that subspace. Finally, the value correspondence is modified as follows:

$$\bar{V}(u) = \begin{cases} V(u) & \text{if } \omega \notin P(u) \\ \{z \in V(u) | z_i \in A_i^0(u_i) \forall i\} & \text{otherwise} \end{cases}$$

Proposition 8: The set of equilibrium allocations of $(\bar{U}, \bar{V})$ for u^0 coincides with the set of Walrasian allocations of u^0 .

Proof: One can prove, as in Lemma 1, that $(u, z) \in \bar{U} \times Z$ with $z \in \bar{V}(u)$ is an equilibrium of $(\bar{U}, \bar{V})$ for u^0 only if $\omega \in P(u)$, which implies that, for each i , $z_i \in A_i^0(u_i)$. Note that $z \in P(u)$ and that there is a unique price p supporting z in u . These two facts imply that the offer surface $h_i(u_{-i})$ faced by each agent i coincides in some open neighborhood N_i of z_i with the hyperplane through z_i normal to p . By an argument analogous to the argument of Proposition 1, we conclude that in N_i , there is no consumption z'_i of $h_i(u_{-i})$ such that $u_i^0(z'_i) > u_i(z_i)$. This means that agent i maximizes u_i^0 over $B_i(p) \cap N_i$ and, by convexity of preferences, it follows that z_i maximizes u_i^0 over all of $B_i(p)$. Q.E.D.

6 Concluding Comments

6.1 Implementation of the Value Correspondence

This paper shows that the value is not immune to manipulation: one should not hope to reach the true value allocations if agents behave strategically. Then, the question arises whether these allocations could be obtained as equilibrium allocations of some

other game. The results concerning the Walrasian correspondence obtained as a by-product of the foregoing analysis, motivate this search: the Walrasian mechanism is not manipulation-proof any more than the value; however, games (see Schmeidler 1980; Hurwicz 1979), and quasi-games (See Section 5), exist that yield the true Walrasian allocations, and only them, as equilibrium allocations. Are there games, or quasi-games, similarly generating the true value allocations at equilibrium?

The answer is negative, as follows from a straightforward adaptation of an important theorem due to Maskin (1977). To state this theorem, we first need a definition:

Definition: The correspondence φ defined on a class of economies E is *monotonic* if

$$\forall u, \quad u' \in E, \quad \forall z \in \varphi(u),$$

$$[\forall i, \forall z' \in F, u_i(z_i) \geq u_i(z'_i) \Rightarrow u'_i(z_i) \geq u'_i(z'_i)] \Rightarrow z \in \varphi(u').$$

Theorem (adapted from Maskin): A correspondence defined on E is implementable by a quasi-game only if it is monotonic.

Maskin states his result for games. The proof for quasi-games is the same.

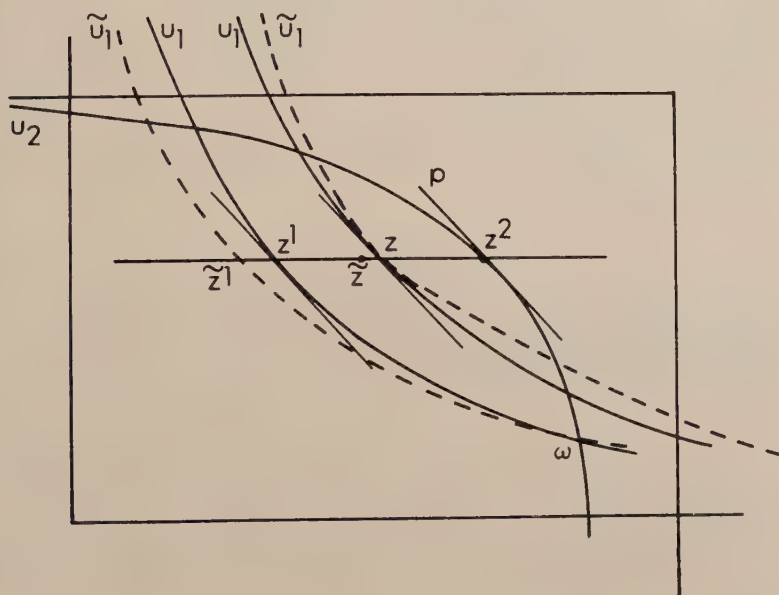


Fig. 9

Proposition 9: There is no quasi-game implementing the value correspondence.

This result follows directly from the above theorem and the following lemma.

Lemma 3: The value correspondence is not monotonic.

Proof: The proof is by way of an example, in which $(l, n) = (2, 2)$, and the notational conventions are as before. The example is represented in Fig. 9. We start from the economy $u = (u_1, u_2)$, which has a unique value allocation $z = (z^1 + z^2)/2$. Then, we replace u_1 by $\tilde{u}_1$ such that agent 1's new lower contour set at z_1 contains his original upper contour set at z_1 , and such that $\tilde{u}_1(z_1^1) > \tilde{u}_1(\omega_1)$. The gains from trade are greater in $(\tilde{u}_1, u_2)$ than in u , and the horizontal distance between $\tilde{z}^1$ and z^2 is greater than the horizontal distance between z^1 and z^2 .

We observe that $\tilde{u}_1$ is obtained from u_1 by a monotonic transformation at z_1 . If V were a monotonic correspondence, z would also be a value allocation of $(\tilde{u}_1, u_2)$. But $V(\tilde{u}_1, u_2)$ is the point $\tilde{z} = (\tilde{z}^1 + z^2)/2$, and $\tilde{z} \neq z$. Q.E.D.

6.2 The Constrained Walrasian Correspondence

Throughout the paper, we have focused on interior allocations. If boundary allocations had been permitted, essentially the same results would have been proved, with the *constrained Walrasian correspondence* CW playing the role of the Walrasian correspondence. CW , introduced by Hurwicz, Maskin and Postlewaite (1979), is defined as W except that maximization of preferences at prices p takes place in the subset of $B_i(p)$ of consumptions z_i satisfying the constraint $z_i \leq \sum \omega_i$. Under convexity of preferences, one of our maintained assumptions, CW and W coincide whenever this constraint is not binding. CW is a monotonic correspondence and it is the smallest monotonic extension of W .

6.3 Implementation of the Walrasian Correspondence in Economies Without Transferable Utility

We saw in Section 5 that, under two alternative and minor respecifications, the value quasi-game implements the Walrasian correspondence. The next proposition says that this implementation property holds even in economies without transferable utility. Note that in this wider class, value allocations are not defined for the true preferences

(we are not concerned here with the so-called NTU value), and again one should see (U, V) as an *abstract* quasi-game (as defined in Section 4).

Proposition 10: Let E' be the class of pure-exchange economies with l commodities and n agents, where agents have preferences representable by continuous, quasiconcave and strictly increasing utility functions. Then, for each $e \in E'$, the sets of equilibrium allocations of the quasi-games (U^s, V) and $(\bar{U}, \bar{V})$, played as abstract quasi-games in E' , coincide with the set of true Walrasian allocations of e .

Proof: The proofs for (U^s, V) and $(\bar{U}, \bar{V})$ are respectively identical to the proofs of Propositions 7 and 8.

6.4 Generalizations

The manipulability of the *Nash bargaining solution* can be analyzed similarly and, in fact, identical conclusions can be established (recall the results obtained by Sobel in the two-person case). The only significant difference concerns the proof of the counterpart of Lemma 1.

Several of the results presented here rely in an essential way on the fact that the value correspondence is a sub-correspondence of the correspondence associating with each economy its set of individually-rational and Pareto-efficient outcomes. In a follow-up paper (Thomson 1984), we provide a general analysis of the class of correspondences that have this property.

6.5

One could think that the close resemblance, under manipulation, of the value and the Walrasian correspondence, is related to the equivalence under truthful behavior of these two correspondences when there are a large number of consumers. However, as also discussed in Thomson (1984), the same phenomenon occurs for correspondences for which this asymptotic result does not hold.

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Deception Games

By V. J. Baston and F. A. Bostock¹

1 Introduction

A general Deception Game $\Gamma(n, k, P_1, \dots, P_n)$ is a two person zero sum game of the following form.

Each component x_i of an ordered n -tuple $(x_1, \dots, x_n)$ of real numbers is chosen independently from the closed interval $[0, 1]$ by means of a probability function P_i on $[0, 1]$. Only player 2 sees the n -tuple and, after seeing it, he presents an n -tuple $(y_1, y_2, \dots, y_n)$ to player 1 which satisfies $y_i = x_i$ for at least $k \leq n$ values of i ; this is the “deception move”. After looking at $(y_1, y_2, \dots, y_n)$ player 1 nominates a positive integer i between 1 and n inclusive and subsequently receives a payoff x_i .

Mark Thompson initiated the study of deception games in an undergraduate thesis at Harvard University. He considered the case when $n = 3, k = 2$ and

- (i) each P_i is the uniform distribution,
- (ii) for $p = 1, 2, 3, 4$ each P_i is the distribution which assigns a probability $1/(p + 1)$ at each of the points $0, 1/p, 2/p, \dots, 1$.

Spencer [1] reveals that a number of deception games have come under investigation but we have been unable to trace any other papers on the subject. Although the game appears intuitively clear, theoretical difficulties can in fact arise concerning the strategies available to the players. Notice that a strategy for player 1 can be thought of as a function from $[0, 1]^n$ to $\{1, 2, \dots, n\}$ and a strategy for player 2 as a function from $[0, 1]^n$ to $[0, 1]^n$ which leaves at least k of the components unchanged. Thus, given the Axiom of Choice, pathological functions could be available to the players, causing uncertainties regarding the resultant expectation. We shall therefore restrict the strategies of the players to be Lebesgue measurable functions.

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Spencer [1] has pointed out that the usual compactness arguments on the strategy space do not seem to establish the existence of a value for the game given by case (i) above. In this note we show that the game $\Gamma(n, k, P_1, \dots, P_n)$ has a value by considering a sequence of corresponding finite games.

2 Preliminary Concepts and Notation

Let u be a fixed positive integer, $P_i (i = 1, \dots, n)$ distribution functions on $[0, 1]$ and $U = \{0, 1/u, \dots, (u-1)/u\}$. For $x \in U$ let

$$I(x) = \begin{cases} [x, x + u^{-1}) & \text{for } x \neq 1 - u^{-1} \\ [1 - u^{-1}, 1] & \text{when } x = 1 - u^{-1} \end{cases}$$

and

$$\delta(P_i, I(x)) = \begin{cases} P_i^-(x + u^{-1}) - P_i^-(x) & \text{if } x \neq 1 - u^{-1} \\ 1 - P_i^-(x) & \text{if } x = 1 - u^{-1} \end{cases}$$

where $P_i^-(t)$ denotes $\lim_{\alpha \rightarrow t^-} P_i(\alpha)$ when $t > 0$ and $P_i^-(0) = 0$.

Let D_i be the distribution on $[0, 1]$ which assigns a probability $\delta(P_i, I(x))$ at the point $x \in U$. Notice that the game $\Gamma_u = \Gamma(n, k, D_1, \dots, D_n)$ is finite; we will use optimal strategies in this game to define strategies in $\Gamma = \Gamma(n, k, P_1, \dots, P_n)$ which will be "close" to optimal when u is large. Denote $[0, 1]^n$ by C and let $C(q) = \bigcup_{i=1}^n I(q_i)$ where $q \in U^n$ so that $\{C(q) : q \in U^n\}$ is a partition of C . For $x \in C$ we can define $w(x)$ as the unique $q \in U^n$ such that $x \in C(q)$; w will be used to send n -tuples in Γ to n -tuples in Γ_u , enabling the players to take advantage of a knowledge of their optimal strategies in Γ_u when playing Γ .

In Γ_u a (mixed) strategy for player 1 gives the probability with which he selects the integer i when faced with an n -tuple $q \in U^n$ and so can be regarded as a function F_u from U^n to C such that $\sum_{i=1}^n (F_u)_i(q) = 1$ for all $q \in U^n$ where $(F_u)_i$ denotes the i -th projection. (Except in Lemma 1 and its proof we will employ a suffix on strategies in Γ_u to make them easily distinguishable from strategies in Γ .) A (mixed) strategy for player 2 in Γ_u gives the probability with which he changes $x \in U^n$ to $y \in U^n$;

notice that if x and y differ in more than $n - k$ components then the probability must be zero. Thus a strategy for player 2 in Γ_u can be regarded as a function G_u from $U^n \times U^n$ to $[0, 1]$ such that $G_u(x, y) = 0$ when $x_i \neq y_i$ for more than $n - k$ values of i , and $\sum_{y \in U^n} G_u(x, y) = 1$ for all $x \in U^n$.

From now on $\sum_p$ and $\sum_q$ will denote sums over U^n and, for a measurable set X , we will write $S(X)$ for $\int_X dP$ where dP stands for $dP_1, \dots, dP_n$.

3 Proof of the Main Result

In Lemma 4 we prove that the value of Γ is the limit of the values of the corresponding games Γ_u as $u \rightarrow \infty$. Thus, in Lemmas 2 and 3, we will effectively be showing that the players can obtain ϵ -optimal strategies in Γ by amending, in a natural way, optimal strategies in some Γ_u . In proving Lemma 2 we shall require the following lemma.

Lemma 1: In Γ_u there is an optimal strategy G for player 2 such that $G(p, q) = 0$ when $\prod_{i=1}^n D_i(q_i) = 0$.

Proof: Suppose the result is false. Let $Q = \left\{ q \in U : \prod_{i=1}^n D_i(q_i) = 0 \right\}$ then, for every optimal strategy G , $T(G) = \{(p, q) : G(p, q) > 0 \text{ and } q \in Q\}$ is non-empty. Choose an optimal strategy H such that $|T(H)|$ equals the minimum of $|T(G)|$ over all optimal strategies G , where $|X|$ denotes the cardinality of X . Choose $(a, b) \in T(H)$ and let $K = \{k : D_k(b_k) = 0\}$. For $k \in K$ we can choose $\beta_k \in U$ such that $D_k(\beta_k) > 0$. Let b' be defined by $b'_i = b_i$ for $i \notin K$, $b'_i = \beta_i$ for $i \in K$ and H' by $H'(a, b) = 0$, $H'(a, b') = H(a, b') + H(a, b)$ and $H'(p, q) = H(p, q)$ otherwise. By the choice of H , H' cannot be optimal so there is a pure strategy F such that the expectation $E_u(F, H') > v_u$, where v_u is the value of the game Γ_u , i.e.

$$\sum_p \prod_{i=1}^n D_i(p_i) \sum_q H'(p, q) p_{F(q)} > v_u. \quad (1)$$

Define the strategy L by $L(b) = F(b')$ and $L(q) = F(q)$ otherwise, then (1) becomes $\sum_p \prod_{i=1}^n D_i(p_i) \sum_q H(p, q) p_{L(q)} > v_u$, i.e.

$$E_u(L, H) > v_u.$$

This proves the result since it contradicts that H is an optimal strategy.

Lemma 2: In the game Γ there exists a strategy G for player 2 such that for every pure strategy F for player 1 the expectation $E(F, G)$ exists and satisfies $E(F, G) \leq v_u + u^{-1}$ where v_u is the value of the game Γ_u .

Proof: Let F be a pure strategy for player 1 in Γ and F_u the strategy for player 1 in Γ_u whereby, when he sees q , he chooses the integer i with probability $S(F^{-1}(i) \cap C(q))/S(C(q))$ if $S(C(q)) \neq 0$ and with probability $1/n$ if $S(C(q)) = 0$. By Lemma 1 we can let G_u be an optimal strategy for player 2 in Γ_u such that $G_u(p, q) = 0$ when $S(C(q)) = \prod_{i=1}^n D_i(q_i) = 0$.

We then have

$$E_u(F_u, G_u) = \sum_p \sum'_q \sum_{i=1}^n S(C(p)) G_u(p, q) p_i S(F^{-1}(i) \cap C(q)) / S(C(q)) \leq v_u \quad (2)$$

where $\sum'_q$ denotes the sum over $q \in U$ for which $S(C(q)) \neq 0$.

In Γ let player 2 adopt the strategy G whereby, when he is faced by $x \in C$, he chooses $y \in C$ as follows. He first chooses $q \in U^n$ with probability $G_u(w(x), q)$ and lets $L(x, q) = \{i : (w(x))_i = q_i\}$ and $J(x, q) = \{i : (w(x))_i \neq q_i\}$. For $i \in L(x, q)$ he takes $y_i = x_i$ whilst for $i \in J(x, q)$ he chooses y_i in $I(q_i)$ by means of the distribution function H given by $H(q_i) = 0$ and $H(x) = (P_i(x) - P_i^-(q_i)) / \delta(P_i, I(q_i))$ otherwise.

Hence

$$\begin{aligned} E(F, G) &= \sum_p \sum'_q G_u(p, q) \left\{ \sum_{i \in L(p, q)} S(C(p)) \int_{F^{-1}(i) \cap C(q)} x_i dP \right. \\ &\quad \left. + \sum_{j \in J(p, q)} \left(\int_{C(p)} x_j dP \right) S(F^{-1}(j) \cap C(q)) \right\} / S(C(q)). \\ &\leq \sum_p \sum'_q \sum_{i=1}^n S(C(p)) G_u(p, q) (p_i + u^{-1}) S(F^{-1}(i) \cap C(q)) / S(C(q)) \\ &\leq v_u + u^{-1} \quad \text{by (2)} \end{aligned}$$

and the proof is complete.

Lemma 3: In the game Γ there exists a strategy F for player 1 such that, for every pure strategy G for player 2, the expectation $E(F, G)$ exists and satisfies $E(F, G) \geq v_u$.

Proof: Let G be a pure strategy for player 2 in Γ and G_u the strategy for player 2 in Γ_u given by

$$G_u(p, q) = S(C(p) \cap G^{-1}(C(q))) / S(C(p)) \quad \text{if } S(C(p)) \neq 0$$

and

$$G_u(p, q) = \begin{cases} 1 & \text{if } q = p \\ 0 & \text{if } q \neq p \end{cases} \quad \text{when } S(C(p)) = 0.$$

(Note that when $q_i \neq p_i$ for more than $n - k$ values of i $C(p) \cap G^{-1}(C(q))$ is empty so $G_u(p, q) = 0$ and further that $\bigcup_q (C(p) \cap G^{-1}(C(q))) = C(p)$ so $\sum_q G_u(p, q) = 1$).

Let F_u be an optimal strategy for player 1 in Γ_u , then

$$E_u(F_u, G_u) = \sum_p S(C(p)) \sum_q G_u(p, q) \sum_{i=1}^n p_i(F_u)_i(q) \geq v_u. \quad (3)$$

Let $H = F_u \circ w$; since $\sum_{i=1}^n H_i(x) = 1$ for all $x \in C$, $H_i(x)$ can be thought of as a probability for player 1 of choosing i when faced with x so that H represents a strategy for player 1 in Γ . Hence

$$E(H, G) = \int_C \left(\sum_{i=1}^n x_i H_i(G(x)) \right) dP$$

and it follows that

$$\begin{aligned} E(H, G) &\geq \sum_p \sum_q \sum_{i=1}^n S(C(p) \cap G^{-1}(C(q))) p_i(F_u)_i(q) \\ &= v_u \quad \text{by (3)} \end{aligned}$$

and the proof is complete.

We now prove our main result in the course of which we show that the sequence v_u has a limit.

Theorem: The game Γ has a value v satisfying $v = \lim_{u \rightarrow \infty} v_u$, where v_u is the value of the corresponding game Γ_u .

Proof: Let $\limsup v_u = a$, $\liminf v_u = b$ and suppose $d = a - b > 0$. Let N be an integer such that $N > 6/d$, then there exist integers $s, t > N$ such that $v_s > a - (d/6)$ and $v_t < b + (d/6)$. By Lemma 2 player 2 can restrict player 1's expectation to at most $v_t + t^{-1} < b + (d/6) + N^{-1} < b + (d/3)$.

Further by Lemma 3 player 1 can ensure an expectation of at least $v_s > a - (d/6) = b + (5d/6)$ which is a contradiction. Thus the sequence (v_u) tends to a limit, v say.

Let $\epsilon > 0$ be given. In Γ player 2 can restrict player 1 to at most $v + \epsilon$ by Lemma 3, whilst player 1 can ensure himself of at least $v - \epsilon$ by Lemma 2. The theorem now follows.

For the special game where $n = 3$, $k = 2$ and each P_i is the uniform distribution, Spencer [1] poses what he calls the Big Question, namely is its value $1/2$? Our paper does not answer this question but does show that it is equivalent to the following. "Does the corresponding game Γ_u always have value $1/2$?"

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Neutral Veto Correspondences with a Continuum of Alternatives¹

By J. Abdou²

Abstract: A veto theory is constructed for a finite committee and a measure space of alternatives. Strategical and descriptive concepts of veto and effectivity functions are extended to the model. We show that under some natural assumptions a neutral veto correspondence can be represented by a real veto power function. As in the discrete case we prove that a stable VC is well-behaved from above and from below, that a convex VC is stable, and that a maximal VC is stable if and only if it is superadditive and subadditive.

1 Introduction

Let N be a society (a committee, a set of players). Given some sociological (economical, political) context, outcomes can be viewed as the results of all possible interactive relations between members of the society. We suppose that these outcomes constitute a well-defined set A . From a descriptive viewpoint, one need not know precisely the strategy-set of each individual leading to the interaction (which includes cooperation and conflict). However repetitions of the activity, structural distribution of skill and power or other objective endowments determine unambiguously for each coalition T acting cooperatively which sets of issues it can block, regardless of the action of other individuals or coalitions. This collection of subsets, denoted $V(T)$, will be called the *veto-set* of the coalition T . A *veto correspondence* (in short VC) associates its veto-set $V(T)$ to each coalition T .

One may also consider those subsets where each coalition T can force the issue of interaction, namely the collection of complementary subsets of $V(T)$. This collection, will be denoted $E(T)$ and the correspondence $T \rightarrow E(T)$ is an *effectivity function* (in

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short EF). The two approaches are essentially the same. Although many authors prefer the EF concept, this paper adopts the VC point of view. As we shall focus our study on additivity notions, we hope that our choice will be partially justified.

A VC (or an EF) can be viewed as the natural generalization of simple games (Nakamura 1975; Abdou 1982). Indeed, a winning coalition can block any proper subset of issues set A and a losing coalition can block none. But recently there has been associated to any game form (g.f.) or any social choice correspondence (s.c.c.) two kinds of EFs, namely, α and β -effectivity functions. Conversely one can ask if for a given EF some game form or some social choice correspondence can be associated and how. (For implementation of simple games see Peleg 1978 or Ishikawa and Nakamura 1978.) Moulin and Peleg (1982) proved that when the g.f. implements a s.c.c. via strong-equilibrium then the α and β -effectivity functions derived from it are the same; that this EF is stable and maximal and that the converse is only true for partial implementation.

Stability is the main strategical concept when a VC is given. For a given profile (collection of preference ordering for each individual) an issue a_0 is *dominated* if some coalition T can improve the utility of all its members by blocking a subset B containing a_0 . The *core* is the set of undominated issues. When an outcome is in the core no coalition objects by blocking it. A VC is *stable* on a preference domain if its core is non-empty for all profiles of the domain. When we restrict ourselves to simple games the Nakamura theorem (Nakamura 1975) provides a necessary and sufficient condition for stability. (See also Ishikawa and Nakamura 1979.)

From a normative point of view one would like to fill in the gaps of the majority principle (existence of Condorcet Paradox). Simple games provide a little ethical improvement as the majority dictatorship is replaced by a winning structure. But the latter is still a dictatorship because it gives decisive power to a winning coalition and no power at all to a losing one.

The minority principle (Moulin 1981, 1982) assumes that all members of society are important and therefore dispose of some blocking power. It follows from this consideration that all coalitions, even small ones, can raise objections to a proposed outcome. The ideal model of the minority principle is a uniform distribution of blocking power in a society (from a normative point of view one would rather say committee): this is *proportional* veto power. Moulin (1982) proved that a *proportional veto function* (PVF) is stable, and that an anonymous and neutral veto correspondence is stable if and only if it is bounded above by the PVF.

More recently Peleg (1982) defined a concept of convexity for effectivity functions. (To see how this concept is related to convexity of games without side payments see Peleg (1982). One can also compare the definition of convexity of characteristic games Shapley (1971), with our results in Section 3.) It follows from his study that a convex VC is stable. His proof relies on a result on cores of ordinal games (see Vilkov 1977, or more directly Greenberg 1985). The nice result however

is restricted to *maximal* VCs: A maximal VC is stable if and only if it is convex or put in another way, if and only if it is both superadditive and subadditive.

It is worth specifying that all the exposed results have been stated when there are finitely many players and issues. In this context the essential remark (which is not merely technical) is that even if we restrict ourselves to maximal VCs, there is no identity between additive VCs and "quasi-additive" VCs, i.e. both superadditive and subadditive VCs. This difference will be pointed out as the "atomic effect": indeed if every single issue (an atom) has no effect when added or removed from a given subset, then the present study shows that the two concepts coincide.

The purpose of this paper is to construct a veto theory over an infinite issue set (a continuum) which presents the same essential features as the finite theory and which enables us to extend most of the interesting results. The main restriction of our model is that we deal only with those VCs which do not distinguish between two alternative subsets having the same measure. Those are μ -neutral VCs where μ is some fixed measure on the alternative set. Moreover we assume that μ is nonatomic so that a real representation of the veto power will be possible.

The study is organized as follows. Section I is a brief survey of general concepts and results already known for the finite case. Section II is devoted to the construction of a model of veto theory over a continuum of alternatives. We prove that a μ -neutral VC can be represented by a real veto power function (Proposition 2.5) and that descriptive concepts on VCs can be translated into similar concepts for their real representations (Proposition 2.8 and 2.11). In Section III we prove that additive VCs are stable and that stable VCs are well behaved from above and from below (Theorem 3.4). Stability of convex VCs is then proved by a direct method (Theorem 3.6). Some proofs are left to the Appendix.

Section I

For any set X , $P(X)$ will denote the set of all subsets of X . The cardinal of X will be written $|X|$. By an algebra $\mathcal{X}$ on X we mean a subset of $P(X)$ which contains the empty set and which is closed under finite union and complementation operations. An algebra $\mathcal{X}$ on X is a σ -algebra if $\mathcal{X}$ is closed under countable union. In this study we suppose given a finite set N called the *committee* or the *set of players*, and an algebra $\mathcal{C}$ on N called the *coalition structure*. We suppose also that a set A is given ($|A| \geq 2$), which may be infinite, called the *alternative set* or the *issue set*, together with an algebra $\mathcal{M}$ on A which is called the *feasible sets structure*. The couple $[(N, \mathcal{C}), (A, \mathcal{M})]$ is called a *veto structure* or an *effectivity structure*.

1.1 Definition: Let $[(N, C), (A, M)]$ be a veto structure, a *veto correspondence* (in short VC) of (N, C) into (A, M) is a mapping

$$V: C \rightarrow P(M)$$

satisfying:

$$H1) \quad V(\emptyset) = \emptyset$$

$$H2) \quad \forall T \in C - \{\emptyset\} \quad A \notin V(T).$$

The interpretation of a VC in the framework of social choice theory is the following: $B \in V(T)$ if and only if coalition T can block the subset B , or set in another way, $B \in V(T)$ if and only if T can force the issue in the complementary $A - B$ of B . For that reason we make the following definition.

1.2 Relation to effectivity functions: Let V be a VC. The pseudo-effectivity function associated with V is the mapping

$$E: C \rightarrow P(M)$$

defined by

$$B \in E(T) \quad \text{iff} \quad A - B \in V(T).$$

All our study could be formulated in "effectivity" terms as well as in "veto" terms. Since our aim is to focus on additivity relations we have taken the second approach in this study.

1.3 A fundamental example: additive VC: We suppose that both N and A are finite sets, that $C = P(N)$ and $M = P(A)$. Let l and m be two positive probability measures on N and A respectively (by positive probability we mean that for each $i \in N$ and each $a \in A$ $l(\{i\}) > 0$ and $m(\{a\}) > 0$). We define the mapping

$$V_{(l,m)}: P(N) \rightarrow P(P(A))$$

by $\forall T \in P(N)$, $V_{(l,m)}(T) = \{B \in P(A) / m(B) < l(T)\}$. Then $V_{(l,m)}$ is a VC. Such a VC is called an *additive veto correspondence*. Given an additive VC, the couple (l, m) such that $V = V_{(l,m)}$ is not unique.

1.4 Classes of preferences: Let $\mathcal{R}$ be the set of all weak orders on A ($R \in \mathcal{R}$ iff R is reflexive, transitive and total), and let $\mathcal{L}$ be the subset of $\mathcal{R}$ constituted of those weak orders which are antisymmetric. For any $R \in \mathcal{R}$ denote by $\hat{R}$ the strict partial order defined by

$$\forall (x, y) \in A \times A : x \hat{R} y \quad \text{iff } x R y \text{ and not } y R x.$$

1.5 Strategic definitions. Core of a VC, and stability of a VC: A profile is a mapping $R : N \rightarrow \mathcal{R}$ which associates a preference order $R^i \in \mathcal{R}$ to any player i . Given a VC V and a profile R , an issue $a \in A$ is *blocked by the coalition* $T \in \mathcal{C}$ if there exists $B \in V(T)$ such that

$$\left\{ \begin{array}{l} a \in B \\ \text{and} \\ \forall i \in T, \quad \forall b \in A - B \quad b \hat{R}^i a. \end{array} \right.$$

$a \in A$ is *dominated* if a is blocked by some coalition T . The *core* $C(V, R)$ of V for the profile R is the set of undominated issues of A .

Let $\mathcal{U} \subset \mathcal{R}$ be some subclass of weak orders on A and let V be a VC. V is *stable* on $\mathcal{U}$ if the core $C(V, R)$ is nonempty for $R \in \mathcal{U}^N$. One can prove that stability on $\mathcal{L}$ is equivalent to stability on $\mathcal{R}$ (at least for a finite alternative set), so that, in the remaining part of this section, by “stability” we mean “stability on $\mathcal{L}$ ” or “stability on $\mathcal{R}$ ”. The following result is due to Moulin and Peleg (1982).

1.6 Theorem: An additive VC is stable.

The converse is not true. A counter-example can be found in Abdou (1981), (Remark 29).

In order to determine the structural features of a stable veto correspondence we need some definitions. The first is due to Peleg (1982); the others can be found in Abdou (1981).

1.7 *Definitions:* Let $(N, C), (A, M)$ be a veto structure.

A veto correspondence V is *convex* if $\forall B_1 \in M, \forall B_2 \in M, \forall T_1 \in C$, and $\forall T_2 \in C$

$$[B_1 \in V(T_1) \text{ and } B_2 \in V(T_2)] \Rightarrow B_1 \cap B_2 \in V(T_1 \cap T_2) \quad \text{or}$$

$$B_1 \cup B_2 \in V(T_1 \cup T_2).$$

V is *well-behaved from below* if: For any finite collection $(T_i, B_i) \ i = 1, \dots, r$ satisfying

$$\begin{cases} T_i \in C, & B_i \in M, & B_i \in V(T_i), & i = 1, \dots, r \\ T_i \cap T_j = \emptyset & i, j = 1, \dots, r, & i \neq j \end{cases}$$

we have $\bigcup_{i=1}^r B_i \neq A$.

V is *well-behaved from above* if: For any finite collection $(T_i, B_i) \ i = 1, \dots, r$ satisfying

$$\begin{cases} B_i \in M, & T_i \in C, & B_i \in V(T_i), & i = 1, \dots, r \\ B_i \cup B_j = A & i, j = 1, \dots, r, & i \neq j \end{cases}$$

we have $\bigcup_{i=1}^r T_i \neq \emptyset$.

The theorems in the present section have been proved under the restrictions that A is finite, $C = P(N)$ and $M = P(A)$. They are actually true for any finite veto structure.

1.8 *Theorem* (Abdou 1981): A stable veto correspondence is well-behaved from below and from above.

1.9 *Theorem* (Peleg 1982): A convex veto correspondence is stable.

The converse is not true. A counter example is given in Peleg (1982).

In the study of implementation of effectivity functions with strong equilibrium a special class of effectivity functions – and then of veto correspondences – arises. These are maximal veto correspondences. Concepts of super and subadditivity are crucial refinements of those of good behavior from below and from above.

1.10 Definitions: Let $(N, C), (A, M)$ be a given veto structure.

A veto correspondence V is *maximal* if $\forall T \in C - \{N\}, \forall B \in M - \{\emptyset\}$

$$B \notin V(T) \Rightarrow A - B \in V(N - T).$$

V is *superadditive* if $\forall T_1 \in C, \forall T_2 \in C, \forall B_1 \in M, \forall B_2 \in M$

$$[T_1 \cap T_2 = \emptyset, B_1 \in V(T_1) \text{ and } B_2 \in V(T_2)] \Rightarrow B_1 \cup B_2 \in V(T_1 \cup T_2)$$

V is *γ -subadditive* if $\forall T_1 \in C, \forall T_2 \in C, \forall B_1 \in M, \forall B_2 \in M$

$$[B_1 \cap B_2 = \emptyset, B_1 \cup B_2 \in V(T_1 \cup T_2)] \Rightarrow B_1 \in V(T_1) \text{ or } B_2 \in V(T_2)$$

V is *δ -subadditive* if $\forall T_1 \in C, \forall T_2 \in C, \forall B_1 \in M, \forall B_2 \in M$

$$[B_1 \cup B_2 = A, B_1 \in V(T_1), B_2 \in V(T_2)] \Rightarrow B_1 \cap B_2 \in V(T_1 \cap T_2).$$

In the following theorem we gather the main results concerning stability of veto correspondences with a finite veto structure. For proofs we refer the reader to Abdou (1981 and 1982), Moulin (1983), and Peleg (1984).

1.11 Theorem: 1) Let V be a veto correspondence, then each of the following statements implies its successor.

a) V is superadditive and γ -subadditive.

b) V is convex.

c) V is stable.

d) V is well-behaved from below and from above.

2) The same claim is true if statement (c) is replaced by (c')

c') V is superadditive and δ -subadditive.

3) If moreover V is maximal then all these statements are equivalent.

Section II. The continuous model

In this section, our aim is to construct a model of veto theory with an infinite alternative set. Although our model is not the most general possible, it will be sufficient to extend most of the results stated in the previous section for the finite case.

We begin by the description of the veto structure: The coalition structure (N, C) is finite. C is any algebra on N . The feasible sets structure (A, M) is as follows: (A, M, μ) is a measure space. μ is a nonatomic probability measure on the σ -algebra M .

All the veto correspondences that we shall introduce throughout this paper will be defined on the veto structure $[(N, C), (A, M)]$.

2.1 A fundamental example: The valued VC.

Let F be the set of mappings $l: C \rightarrow [0, 1]$ such that $l(\emptyset) = 0$. For every $l \in F$ we define the following correspondence $V_l: C \rightarrow P(M)$ by

$$V_l(T) = \{B \in M / \mu(B) < l(T)\} \quad (\forall T \in C).$$

Clearly V_l is a VC. Such a VC is called a *valued VC*.

Since our aim is the study of the μ -neutral VCs, we give some definitions.

2.2 Definitions: A VC is μ -neutral if $\forall T \in C, \forall B \in M, \forall B' \in M$

$$[\mu(B) = \mu(B') \text{ and } B \in V(T)] \Rightarrow B' \in V(T)$$

V is M -monotonic if $\forall T \in C, \forall B \in M, \forall B' \in M$

$$[B' \subset B \text{ and } B \in V(T)] \Rightarrow B' \in V(T)$$

V is C -monotonic if $\forall T \in C, \forall T' \in C, \forall B \in M$

$$[T \subset T' \text{ and } B \in V(T)] \Rightarrow B \in V(T')$$

Let V_l be a valued VC. It is clear that V_l is μ -neutral and M -monotonic.

It is natural to ask what is the "topological" property that makes the inequality be strict in the definition of the valued VC. The answer needs the following.

2.3 Definition: A VC V is σ -open if for any coalition $T \in \mathcal{C}$ and any sequence $(B_q)_{q=1,2,\dots}$ satisfying

$$B_q \in M \quad q = 1, 2, \dots$$

$$B_{q+1} \subset B_q \quad q = 1, 2, \dots$$

$$B_q \notin V(T) \quad q = 1, 2, \dots$$

we have $\bigcap_{q=0}^{\infty} B_q \notin V(T)$.

Given the veto structure $[(N, \mathcal{C}), (A, M)]$ the set of all μ -neutral, M -monotonic, σ -open VCs will be denoted by $\mathcal{V}$. We can now define what we mean by a veto power.

2.4 Definition: Let $V \in \mathcal{V}$. The *veto power function* (VPF) associated with V is the mapping

$$v: \mathcal{C} \rightarrow [0, 1]$$

defined by

$$v(T) = \begin{cases} \sup \{ \mu(B) / B \in V(T) \} & \text{if } V(T) \neq \emptyset \\ 0 & \text{if } V(T) = \emptyset. \end{cases}$$

Clearly v is an element of $\mathcal{F}$. For a coalition T the real $v(T)$ represents the magnitude of its veto power. The following proposition is a full characterization of elements of $\mathcal{V}$.

2.5 Proposition: The collection $\mathcal{V}$ of all μ -neutral, M -monotonic σ -open VCs is identical to the collection of all valuated VCs $\{V_l | l \in \mathcal{F}\}$. More precisely: To any $V \in \mathcal{V}$, the VPF v associated to V is in $\mathcal{F}$ and is such that

$$V(T) = \{B \in M | \mu(B) < v(T)\} \quad (\forall T \in \mathcal{C}).$$

Conversely, to any $l \in \mathcal{F}$, V_l is in $\mathcal{V}$ and the VPF associated to V_l is l itself.

Proof: Let $V \in \mathcal{V}$ and let v be its associated VPF. If $V(T) = \emptyset$ there is nothing to prove. If $V(T) \neq \emptyset$ we have by definition of v

$$B \in V(T) \Rightarrow \mu(B) \leq v(T).$$

We must show that the inequality $\mu(B) \leq v(T)$ is strict. Assume on the contrary that $\mu(B) = v(T)$. First we rule out the case $v(T) = 1$, otherwise by μ -neutrality $A \in V(T)$ which contradicts assumption H2. If $\mu(B) = v(T) < 1$, we can construct by Lemma 1 (see appendix) a sequence (B_k) of sets satisfying

$$B_k \in \mathcal{M} \quad k = 1, 2, \dots$$

$$B \subset B_{k+1} \subset B_k \quad k = 1, 2, \dots$$

$$\mu(B_k) = \mu(B) + \frac{1 - \mu(B)}{k}, \quad k = 1, 2, \dots$$

Clearly $B_k \notin V(T)$, $k = 1, 2, \dots$. Since V is σ -open $\bigcap_{k=1}^{\infty} B_k \notin V(T)$. Since $\mu\left(\bigcap_{k=1}^{\infty} B_k\right) = \mu(B)$, by μ -neutrality $B \notin V(T)$ which is the desired contradiction.

Now let $B \in \mathcal{M}$ be such that $\mu(B) < v(T)$. By definition of $v(T)$, there exists $B' \in V(T)$ such that

$$0 \leq \mu(B) < \mu(B') \leq v(T) \leq 1.$$

By Lemma 1, there exists $B'' \in \mathcal{M}$ such that

$$B \subset B''$$

$$\mu(B'') = \mu(B').$$

By μ -neutrality, $B'' \in V(T)$. By $\mathcal{M}$ -monotonicity, $B \in V(T)$.

Q.E.D.

2.6 Remark: In the original definition of a VC, one could add the following requirement (see Definition 1.1).

$$\text{H3)} \quad \forall T \in \mathcal{C} - \{\emptyset\} \quad \emptyset \in V(T).$$

If this condition is satisfied then superadditivity (or convexity) implies C -monotonicity. Let $V \in \mathcal{V}$ and let v be its associated VPF, then V satisfies (H3) if and only if

$$v(T) > 0 \quad \text{for all } T \in \mathcal{C} - \{\emptyset\}.$$

Let $\mathcal{V}^*$ be the set of all elements of $\mathcal{V}$ which satisfy (H3) and let $\mathcal{F}^*$ be the corresponding set of VPFs.

We denote by $\mathcal{V}_c$ the set of C -monotonic VCs. Now we give a series of definitions on functions of $\mathcal{F}$ so that all the properties on VCs can be related to similar classical properties on real functions.

2.7 Definitions: Let $l \in \mathcal{F}$.

l is *well-behaved from below* if for every finite collection (T_i) , $i = 1, \dots, r$ such that

$$\begin{cases} T_i \in \mathcal{C} & i = 1, \dots, r \\ T_i \cap T_j = \emptyset & i, j = 1, \dots, r, i \neq j \end{cases}$$

we have $\sum_{i=1}^{i=r} l(T_i) \leq 1$.

l is *well-behaved from above* if for every finite collection (T_i) , $i = 1, \dots, r$, such that

$$\begin{cases} T_i \in \mathcal{C} & i = 1, \dots, r \\ \bigcap_{i=1}^{i=r} T_i = \emptyset \end{cases}$$

we have $\sum_{i=1}^{i=r} (1 - l(T_i)) \geq 1$.

l is *convex* if for every couple $(T_1, T_2) \in \mathcal{C}^2$ we have $l(T_1) + l(T_2) \leq l(T_1 \cap T_2) + l(T_1 \cup T_2)$.

l is *superadditive* if for every couple $(T_1, T_2) \in \mathcal{C}^2$ such that $T_1 \cap T_2 = \emptyset$ we have $l(T_1) + l(T_2) \leq l(T_1 \cup T_2)$.

l is γ -*subadditive* if for every couple $(T_1, T_2) \in \mathcal{C}^2$ we have $l(T_1) + l(T_2) \geq l(T_1 \cup T_2)$.

l is δ -subadditive iff for every couple $(T_1, T_2) \in \mathcal{C}^2$ we have $(1 - l(T_1)) + (1 - l(T_2)) \geq 1 - l(T_1 \cap T_2)$.

l is $\mathcal{C}$ -monotonic iff for any couple $(T_1, T_2) \in \mathcal{C}^2$ we have $T_1 \subset T_2 \Rightarrow l(T_1) \leq l(T_2)$.

As elements of $\mathcal{V}$ and those of $\mathcal{F}$ are described by properties bearing the same names. We have nice conversion statements:

2.8 Proposition:

- 1) Let $V \in \mathcal{V}$ and let $v \in \mathcal{F}$ be its associated VPF then V is well-behaved from below (respectively, well-behaved from above, γ -subadditive, δ -subadditive, $\mathcal{C}$ -monotonic) if and only if v satisfies the same property.
- 2) If v is convex (respectively superadditive) then V satisfies the same property.
- 3) If V is in $\mathcal{V}_c$ or $\mathcal{V}^*$ then the converse of Claim 2 is also true.

Proof: See the Appendix.

2.9 Remark: In the framework of the continuous model studied in this section, there are no maximal VCs. Indeed if V is maximal then $V(N) = M - \{A\}$. Since the measure μ is nonatomic, it follows that $\mu(A - \{x_0\}) = \mu(A) = 1$ for any $x_0 \in A$, which contradicts the μ -neutrality of V .

The following definitions introduce a weaker property than maximality.

2.10 Definitions: Let $l \in \mathcal{F}$.

l is *maximal* if $(\forall T \in \mathcal{C}) l(T) + l(N - T) \geq 1$.

l is *additive* if l is both superadditive and γ -subadditive.

Let $V \in \mathcal{V}$ and let $v \in \mathcal{F}$ be its associated VPF.

V is μ -*maximal* if v is maximal.

V is *additive* if v is additive.

The following proposition can be compared to Claims 2 and 3 of Theorem 1.11.

2.11 Proposition:

- 1) Let $V \in \mathcal{V}$ and let $v \in \mathcal{F}$ be its associated VPF. Each of the following statements implies its successor.
 - a) V is superadditive and γ -subadditive.
 - b) V is convex.
 - c) V is super additive and δ -subadditive.

- d) V is well-behaved from above and from below.
- 2) Claim 1 remains true if we replace V by v .
- 3) If V is μ -maximal, then all these statements are equivalent.

Proof: Claim 1 is identical to Claim 2 of Theorem 1.11. The proof of Claim 2 is straightforward and therefore omitted. Only Claim 3 needs some details.

First argument: Let us call (a') , (b') , (c') and (d') respectively the same statements as (a) , (b) , (c) and (d) , but with V replaced by v . It is easy to see that under maximality of v , statements (a') , (b') , (c') and (d') are equivalent.

Second argument: if V is superadditive (or convex) and μ -maximal then V is in V_C .

Indeed assume, for the sake of contradiction that for some $T_1 \in C$, $T_2 \in C$ we have $T_1 \subset T_2$ and $v(T_2) < v(T_1)$. By Lemma 1, there exists $B \in \mathcal{M}$ such that $v(T_2) < \mu(B) < v(T_1)$. By maximality of v , and measure properties

$$\mu(A - B) = 1 - \mu(B) < 1 - v(T_2) \leq v(N - T_2)$$

so that at the same time

$$B \in V(T_1) \quad \text{and} \quad A - B \in V(N - T_2).$$

Superadditivity (or convexity) of V implies $A = B \cup (A - B) \in V(T_1 \cup (N - T_2))$, a contradiction.

The conclusion of Claim 3 follows now from the construction of the following logical scheme. If V is μ -maximal then

$$(d) \Rightarrow (d') \quad (\text{Prop. 2.8 Claim 1})$$

$$(d') \Rightarrow (c') \quad (\text{first argument})$$

$$(c') \Rightarrow (c) \quad (\text{Prop. 2.8 Claim 1 and 2})$$

$$(c) \Rightarrow (c') \quad (\text{second argument and Prop. 2.8 Claim 3})$$

$$(c') \Rightarrow (b') \quad (\text{first argument})$$

$(b') \Rightarrow (b)$ (Prop. 2.8 Claim 2)

$(b) \Rightarrow (b')$ (second argument and Prop. 2.8 Claim 3)

$(b') \Rightarrow (a')$ (first argument)

$(a') \Rightarrow (a)$ (Prop. 2.8 Claim 1 and 2),

Q.E.D.

The next section deals with strategical properties of VCs.

Section III. Stability of a VC

In the previous section, the feasible set structure (A, M) was assumed to be a measurable space. The neutrality was assumed through the existence of a nonatomic probability measure μ , which provides a "weight" for any measurable alternative subset. Now we introduce a topology on A so that A can be considered as a *compact Hausdorff space*.

Let G be the collection of open sets of A . We require some compatibility between the measure and topology structures. More specifically we assume the following

(R) $G \subset M$ and μ is regular.

The main consequence of the regularity assumption is that for any VC $V \in V$ and any coalition $T \in C$ we have

(C) If $B \in V(T)$, then there exists an open set G such that $B \subset G$ and $G \in V(T)$.

Indeed let v be the VPF associated to V and let $B \in V(T)$. We have $\mu(B) < v(T)$. Put $\epsilon = \frac{v(T) - \mu(B)}{2}$. Since μ is regular and $G \subset M$ there exists an open set G such that $B \subset G$ and $\mu(G - B) < \epsilon$. Clearly $\mu(G) < v(T)$.

Q.E.D.

We recall that R is the collection of all weak orders on A . In the present context we are interested only in those weak orders induced by M -measurable real functions ranging in $[0, 1]$. Let f be such a function. We associate with f the weak order R_f defined

by

$$\forall (a, b) \in A \times A \quad aR_fb \text{ iff } f(a) \geq f(b).$$

The collection of all such weak orders will be denoted by R_1 . When a collection $u = (u_i)_{i \in N}$ of functions is given, we denote indifferently by $C(V, R_u)$ or $C(V, u)$ the core of V for the profile $(R_{u_i})_{i \in N}$.

The following theorem is a direct consequence of the property (C).

3.1 Theorem: Let $V \in \mathcal{V}$ and let $u = (u_i)_{i \in N}$ be a collection of continuous functions on A . Then the core $C(V, u)$ is closed (therefore compact).

Proof: We must show that the dominated set $A - C(V, u)$ is open. Let a be dominated. Then for some $T \in \mathcal{C}$ and some $B \in V(T)$ we have $a \in B$, and $\forall i \in T, \forall b \in A - B, u_i(a) < u_i(b)$.

In view of property (C), there exists an open set G such that $B \subset G \in V(T)$.

Then we have $\forall i \in T, \forall b \in A - G, u_i(a) < u_i(b)$.

Since $A - G$ is compact, for any $i \in T$, there exists $b_i \in A - G$ such that

$$u_i(b_i) = \min_{b \in A - G} u_i(b).$$

By continuity of u_i , there exists an open neighborhood ω_i of a such that $u_i(x) < u_i(b_i)$ ($\forall x \in \omega_i$). Clearly $G \cap [\bigcap_{i \in T} \omega_i]$ is a neighborhood of a and all its elements

are dominated via the same coalition T .

Q.E.D.

In the study of stability one must face two opposite constraints. Let $\mathcal{U}$ be a subset of R_1 . If $\mathcal{U}$ is very large it will be easy to derive any necessary property for stability on $\mathcal{U}$. For instance if $\mathcal{U} = R_1$ the conditions established in the finite case remain trivially true since it is easy to construct suitable profiles to adapt the proofs of Theorem 1.11 to this case. However, we cannot readily show stability even for very well behaved VCs. On the other hand if $\mathcal{U}$ contains only very regular functions it will be easy to show stability for large classes of VCs and impossible to derive any necessary conditions. In this study we limit stability to a special set $R_0 \subset R_1$ as described below.

Let $\mathcal{U}_0$ be the set of continuous functions

$f: A \rightarrow [0, 1]$ such that

the “iso-utility surfaces” have null-measure

$$\forall \lambda \in [0, 1] \mu(\{x \in A / f(x) = \lambda\}) = 0.$$

We denote by $\mathcal{R}_0$ the collection of weak orders induced by $\mathcal{U}_0$. We assume that $\mathcal{R}_0$ is *non empty*. This assumption implies that for any $a \in A$ $\mu(\{a\}) = 0$, but this is a requirement which is easy to satisfy in usual spaces like compacts of R^n with normalized Lebesgue-measures.

Before we study stability on $\mathcal{R}_0$, we wish to show that $\mathcal{R}_0$ have some properties which make it a good candidate for our purpose. We recall that, as a consequence of Theorem 3.1, the core $C(V, R)$ is compact for any $R \in \mathcal{R}_0^N$. That was the first property that we liked to point out. Let V be a VC, the $\mathcal{C}$ -monotonic cover of V is the new VC defined by

$$V_c(T) = \bigcup_{\substack{T' \subset T \\ T' \in \mathcal{C}}} V(T') \quad (\forall T \in \mathcal{C}).$$

It is easy to show that for any profile $R \in \mathcal{R}^N$ $C(V_c, R) = C(V, R)$.

The second desirable property of a subclass of preferences is described in the

3.2 Definition: Let $\mathcal{U} \subset \mathcal{R}_1$ be a subcollection of weak orders. $\mathcal{U}$ is said to be rich if for any couple of VCs V_1 and V_2 we have:

$$(\forall R \in \mathcal{U}^N C(V_1, R) = C(V_2, R)) \Rightarrow V_{1c} = V_{2c}.$$

3.3 Proposition: The collection $\mathcal{R}_0$ is rich.

Proof: Since $C(V, R) = C(V_c, R)$ for any $V \in \mathcal{U}$ and any profile R , it is sufficient to make the proof for any couple of $\mathcal{C}$ -monotonic VCs.

Let V_1 and V_2 be two such VCs and assume that $V_1 \neq V_2$. We seek to construct a profile $\underline{R}$ in $\mathcal{R}_0^N$ s.t.

$$C(V_1, \underline{R}) \neq C(V_2, \underline{R}).$$

Let v_1 and v_2 be the VPFs associated to V_1 and V_2 . There exists $T_1 \in \mathcal{C}$, $T_1 \neq \emptyset$ s.t. $v_1(T_1) \neq v_2(T_1)$. Without loss of generality assume $v_1(T_1) < v_2(T_1)$. Put $\epsilon = \frac{v_2(T_1) - v_1(T_1)}{2}$.

Let f_0 be an arbitrary function in $\mathcal{U}_0$. There exist two real numbers α and β s.t. $\mu\{x/f_0(x) \leq \alpha\} = v_1(T_1)$, $\mu\{x/f_0(x) \leq \beta\} = v_1(T_1) + \epsilon$, and $\{x/f_0(x) = \beta\} \neq \emptyset$. It is clear that: $0 \leq \alpha < \beta < 1$.

Let $\underline{R}$ be the profile induced by the following $\underline{u} \in \mathcal{U}_0^N$.

$$\forall i \in T_1, \quad u_i(x) = f_0(x) \quad \forall x \in A.$$

$$\forall i \in N - T_1, \quad \begin{cases} u_i(x) = f_0(x) & \text{if } f_0(x) \leq \beta. \\ u_i(x) = \beta - \frac{\beta}{1-\beta}(f_0(x) - \beta) & \text{if } f_0(x) > \beta. \end{cases}$$

Let $a \in A$ satisfy $f_0(a) = \beta$ and let $T \in \mathcal{C}$. Put $P_r^c(a, T) = \{x \in A / \exists i \in T u_i(x) \leq u_i(a)\}$. Let $T \in \mathcal{C} - \{\emptyset\}$. If $T \not\subset T_1$ then $P_r^c(a, T) = A$ and a cannot be dominated via T . If $T \subset T_1$ then $P_r^c(a, T) = \{x/f_0(x) \leq \beta\}$ and

$$v_1(T) \leq v_1(T_1) < \mu[P_r^c(a, T)] < v_2(T_1).$$

In conclusion $a \notin C(V_2, \underline{R})$, $a \in C(V_1, \underline{R})$.

We can state now the main theorem of this study. It deals with necessary and sufficient conditions for the stability of VCs.

3.4 Theorem: Let $V \in \mathcal{V}$. Then each of the following statements implies its successor

- a) V is additive.
- b) V is stable on $\mathcal{R}_0$.
- c) V is well-behaved from below and from above.

Moreover if V is μ -maximal then (a), (b), (c) are equivalent.

Proof of $a \Rightarrow b$: We put $N = \{1, 2, \dots, n\}$. Let $\mathcal{C}_0$ be the set of minimal non-empty sets in $\mathcal{C}$ (the “atoms” of $\mathcal{C}$).

$$\mathcal{C}_0 = \{T \in \mathcal{C} - \{\emptyset\} / \forall T' \in \mathcal{C}, T' \subset T \Rightarrow T' = T \text{ or } T' = \emptyset\}.$$

We claim that C_0 is a partition of N . Indeed let T and S be in C_0 . We have either $T \cap S = S = T$ or $T \cap S = \emptyset$. On the other hand $\forall i \in N \exists T_0 \in C_0, i \in T_0$. Then clearly $N = \bigcup_{T \in C_0} T$.

Let $V \in C$ and let v be its associated VPF. For $i \in N$ put $v_i = \frac{v(T)}{|T|}$, where T is the unique set of C_0 containing i .

We have $\forall T \in C \ v(T) = \sum_{i \in T} v_i$. Without loss of generality we can assume that $v_i > 0 (\forall i \in N)$. (Otherwise we restrict the VC to the set $N^* = \{i \in N | v_i > 0\}$ and notice that the core remains unchanged.)

Let $\underline{u} \in \mathcal{U}_0^N$ and let $\underline{R} \in \mathcal{R}_0^N$ be the associated profile to $\underline{u}$. The real function

$$\phi_1 : t \mapsto \mu\{x \in A | u_1(x) < t\}$$

which maps $[0, 1]$ on itself is continuous. $\phi_1(0) = 0$ and $\phi_1(1) = 1 \geq \sum_{i \in N} v_i \geq v_1 > 0$.

Since ϕ_1 is continuous on $[0, 1]$ the set $L_1 = \{t \in [0, 1] | \phi_1(t) = v_1\}$ is non empty. Put $t_1 = \min L_1 > 0$. Put $A_0 = A$.

We have $t_1 \in u_1(A_0)$. Indeed if this were not true, by continuity of u_1 and compacity of A_0 there would exist an interval $]t_1 - \epsilon, t_1 + \epsilon[$ such that

$$]t_1 - \epsilon, t_1 + \epsilon[\cap u_1(A_0) = \emptyset$$

which would contradict that $t_1 = \min L_1$.

Put $A_1 = \{x \in A_0 | u_1(x) \geq t_1\} \neq \emptyset$. If $N = \{1\}$ the construction is already finished. If $|N| \geq 2$, we assume by induction that $(A_1, \dots, A_k, t_1, \dots, t_k)$ for $k \leq n-1$ have been constructed and that they satisfy

$$t_i \in [0, 1] \quad i = 1, \dots, k$$

$$A_i = \{x \in A | \forall j \in \{1, \dots, i\} u_j(x) \geq t_j\} \neq \emptyset \quad i = 1, \dots, k$$

$$\mu(A_i) = 1 - \sum_{j=1}^{j=i} v_j \quad i = 1, \dots, k.$$

The real function $\phi_{k+1} : t \mapsto \mu\{x \in A_k | u_{k+1}(x) < t\}$ which maps $[0, 1]$ on itself is continuous. Since $\phi_k(0) = 0$ and $\phi_k(1) = \mu(A_k) = 1 - \sum_{j=1}^k v_j \geq v_{k+1} > 0$ the set $L_{k+1} = \{t \in [0, 1] | \phi_{k+1}(t) = v_{k+1}\}$ is nonempty. Put $t_{k+1} = \min L_{k+1}$. Clearly $t_{k+1} > 0$. Moreover we have $t_{k+1} \in u_{k+1}(A_k)$. Otherwise by continuity of u_{k+1} and

compactness of A_k ,

$$]t_{k+1} - \epsilon, t_{k+1} + \epsilon[\cap_{u_{k+1}} (A_k) = \emptyset \text{ for some } \epsilon > 0.$$

Put $A_{k+1} = \{x \in A_k \mid u_{k+1}(x) \geq t_{k+1}\} \neq \emptyset$. By induction A_n is constructed. It satisfies

$$A_n = \{x \in A \mid \forall i \in \{1 \dots n\} u_i(x) \geq t_i\} \neq \emptyset$$

$$\mu(A_n) = 1 - \sum_{i=1}^{i=n} v_i \geq 0.$$

We claim that $A_n \subset C(V, \underline{R})$. For any $a \in A$ and $T \in \mathcal{C}$ put

$$\text{Pr}(a, T) = \{x \in A \mid \forall i \in T u_i(a) < u_i(x)\},$$

$$\text{Pr}^c(a, T) = A - \text{Pr}(a, T).$$

We have $\text{Pr}^c(a, T) = \bigcup_{i \in T} \text{Pr}^c(a, \{i\})$.

$$a \in C(V, \underline{R}) \quad \text{iff} \quad \forall T \in \mathcal{C} \quad \text{Pr}^c(a, T) \not\subset V(T).$$

Let us compute $\text{Pr}^c(a, T)$ for $a \in A_n$ and $T \in \mathcal{C}$. For $i \in T$, $A_{i-1} - A_i \subset \text{Pr}^c(a, \{i\})$

$$\mu[\text{Pr}^c(a, T)] = \mu\left[\bigcup_{i \in T} \text{Pr}^c(a, \{i\})\right] \geq \sum_{i \in T} \mu(A_{i-1} - A_i) = \sum_{i \in T} v_i = v(T)$$

Therefore $\text{Pr}^c(a, T) \not\subset V(T)$ and $a \in C(V, \underline{R})$

Q.E.D.

Proof of $b \Rightarrow c$: Suppose that V is not well-behaved from below. In view of Proposition 2.8 statement (1), our assumption is equivalent to the existence of a finite collection $(T_i)_{i=1, \dots, r}$ such that

$$\begin{cases} T_i \in \mathcal{C} & \forall i = 1, \dots, r \\ T_i \cap T_j = \emptyset & \forall i, j = 1, \dots, r, i \neq j, \quad \text{and} \\ \sum_{i=1}^{i=r} v(T_i) > 1. \end{cases}$$

Without loss of generality we can suppose $v(T_i) > 0, i = 1, \dots, r$.

There exists a collection of positives numbers $(\alpha_i), i = 1, \dots, r, \alpha_i > 0, \forall i = 1, \dots, r$ satisfying

$$\sum_{i=1}^{i=r} (v(T_i) - \alpha_i) = 1.$$

$$v(T_i) - \alpha_i > 0 \quad \forall i = 1, \dots, r.$$

Put

$$\gamma_i = \sum_{k=1}^{k=i} (v(T_k) - \alpha_k), \quad i = 1 \dots r, \quad \text{and} \quad \gamma_0 = 0.$$

Since $\mathcal{U}_0 \neq \emptyset$ let f_0 be an arbitrary element of $\mathcal{U}_0$. The real function $\psi : t \rightarrow \mu(\{x / f_0(x) \leq t\})$ is continuous on $[0, 1]$. There exists a collection $(s_i), i = 1 \dots r$, of real numbers satisfying

$$\begin{cases} \psi(s_i) = \gamma_i & i = 1, \dots, r. \\ 0 = s_0 < s_1 < \dots < s_r = 1. \end{cases}$$

Put

$$t_i = \frac{s_i + s_{i-1}}{2} \quad i = 1, \dots, r.$$

$$B_i = \{x \in A / s_{i-1} \leq f_0(x) \leq s_i\} \quad i = 1, \dots, r$$

We have $\bigcup_{i=1}^{i=r} B_i = A$.

Define now the following functions in $\mathcal{U}_0$

$$u_i : A \rightarrow [0, 1], \quad i = 1, \dots, r$$

$$u_i(x) = |f_0(x) - t_i| \quad i = 1, \dots, r,$$

and the profile $\underline{u} = (u_e)_{e \in N}$ such that

$$u_e = u_i \quad \text{iff } e \in T_i \quad i = 1, \dots, r$$

$$u_e = f_0 \quad \text{iff } e \notin \bigcup_{i=1}^{i=r} T_i.$$

Let $\underline{R}$ be the profile in $\mathcal{R}_0$ associated with $\underline{u}$. The reader can easily verify that $C(V, \underline{R}) = \emptyset$.

Assume now that V is not well-behaved from above; by Proposition 2.8, Claim 1 this assumption is equivalent to the existence of a finite collection (T_i) , $i = 1, \dots, r$ satisfying

$$\bigcap_{i=1}^{i=r} T_i = \emptyset \quad \text{and} \quad \sum_{i=1}^{i=r} (1 - v(T_i)) < 1.$$

There exists a collection of real numbers (m_i) , $i = 1, \dots, r$; satisfying

$$m_i > 1 - v(T_i) \quad \forall i = 1, \dots, r \quad \text{and} \quad \sum_{i=1}^{i=r} m_i < 1.$$

Put:

$$\alpha = \frac{1}{r} \left(1 - \sum_{i=1}^{i=r} m_i \right) > 0$$

$$\beta_i = \sum_{k=1}^{k=i} m_k + (i-1)\alpha \quad i = 1, \dots, r$$

$$\gamma_i = \sum_{k=1}^{k=i} m_k + i\alpha \quad i = 1, \dots, r.$$

Let f_0 be an arbitrary element of $\mathcal{U}_0$. Define $\psi : [0, 1] \rightarrow [0, 1]$ as follows: $\psi(t) = \mu(\{x/f_0(x) \leq t\})$. ψ is continuous. There exists a collection (t_i, s_i) , $i = 1 \dots r$ satisfying

$$\psi(t_i) = \beta_i \quad i = 1, \dots, r$$

$$\psi(s_i) = \gamma_i \quad i = 1, \dots, r$$

$$0 < t_1 < s_1 < \dots < t_i < s_i < t_{i+1} < s_{i+1} \dots < t_r < s_r = 1.$$

Put:

$$B_i = \{x \in A / s_{i-1} < f_0(x) < t_i\} \quad i = 2, \dots, r$$

$$B_1 = \{x \in A / f_0(x) < t_1\}$$

$$\tilde{B}_i = \{x \in A / s_{i-1} \leq f_0(x) \leq t_i\} \quad i = 2, \dots, r$$

$$\tilde{B}_1 = \{x \in A / f_0(x) \leq s_1\}.$$

The following functions $\tilde{u}_i$ are in $\mathcal{U}_0$

$$\tilde{u}_1(x) = \begin{cases} f_0(x) & \text{if } f_0(x) \leq t_r \\ \frac{s_r - f_0(x)}{s_r - t_r} (1 - s_r + t_r) & \text{if } f_0(x) \geq t_r \end{cases}$$

$$\tilde{u}_i(x) = \begin{cases} f_0(x) - s_{i-1} & \text{if } f_0(x) \geq s_{i-1} \\ f_0(x) + (1 - s_{i-1}) & \text{if } f_0(x) \leq t_{i-1} \\ \frac{s_{i-1} - f_0(x)}{s_{i-1} - t_{i-1}} (1 - s_{i-1} + t_{i-1}) & \text{if } t_{i-1} \leq f_0(x) \leq s_{i-1}. \end{cases}$$

To define now a profile in R_0 we notice that $\bigcup_{i=1}^{i=r} (N - T_i) = N$. There exists a map $\pi : N \rightarrow \{1 \dots r\}$ such that $\pi(e) = k \Rightarrow e \in N - T_k$.

Let $\underline{u} = (u_e)_{e \in N}$ the profile where $u_e = \tilde{u}_i$ iff $\pi(e) = i$. We now identify $\{1, \dots, r\}$ with the quotient set $\mathbb{Z} / r\mathbb{Z}$ with the additive group structure.

For any $x \in A$ and any $T \in \mathcal{C}$ put

$$\text{Pr}(x, T) = \{y \in A / \forall i \in T u_i(x) < u_i(y)\} \quad \text{and}$$

$$\text{Pr}^c(x, T) = A - P_r(x, T).$$

Let $x \in A$. There exists $i \in \mathbb{Z} / r \mathbb{Z}$ such that $x \in \tilde{B}_i$. We claim that $B_{i+1} \subset \text{Pr}(x, T_{i+1})$. Indeed, if $e \in T_{i+1}$ then $\pi(e) \neq i+1$, so that the definition of the profile yields

$$\forall y \in B_{i+1} \quad u_e(x) < u_e(y).$$

Hence our claim is true.

Since $\mu(B_{i+1}) = m_{i+1}$ and $\text{Pr}^c(x, T_{i+1}) \subset B_{i+1}^c$ we have

$$\mu[\text{Pr}^c(x, T_{i+1})] \leq \mu(B_{i+1}^c) = 1 - \mu(B_{i+1}) = 1 - m_{i+1} < v(T_{i+1}).$$

Hence x is dominated via T_{i+1} . Since x is arbitrary we have proved that $C(V, \underline{u}) = \emptyset$.

The second part of the claim follows from Theorem 2.11, Claim 3.

3.5 Remark: Without the μ -maximality assumption, properties (a), (b), (c) of Theorem 3.4 are not equivalent as we can verify with the following examples.

Let A be the real interval $[0, 1]$, $\mathcal{M}$ be the Borel σ -algebra on A and μ be the Lebesgue-measure.

1) If $|N| = 3$, and V has the following VPF: $v(T) = \frac{1}{3}$ if $|T| = 1$, $v(T) = \frac{1}{3}$ if $|T| = 2$, and $v(N) = 1$. It can be shown that for such a VC (b) holds but not (a).

2) If $|N| = 5$, and V has the following VPF: $v(T) = 0$ if $|T| = 1$; $v(T) = \frac{1}{2}$ if $|T| = 2$, $|T| = 3$ or $|T| = 4$; and $v(N) = 1$. By Theorem 1.2 of Abdou (1981), property (b) does not hold for V , but it can be verified that property (c) does.

With very little effort we can state a theorem about stability of convex VCs. Actually in our context, the following theorem is a consequence of Theorem 3.4 and properties of convex real valued functions.

3.6 Theorem: Let $V \in \mathcal{V}$. If V is convex then V is stable on $\mathcal{R}_0$. Moreover if V is μ -maximal then the converse is also true.

Proof: If V is convex then V is weaker than some additive VC defined on $\mathcal{C}$. More precisely let v be the VPF associated to V , then for some additive $l \in \mathcal{F}$ we have

$$v(T) \leq l(T) \quad \forall T \in \mathcal{C}.$$

Indeed, let C_0 be the set of "atoms" of C (see the proof of Theorem 3.4 above). We put $C_0 = \{\alpha_1, \alpha_2, \dots, \alpha_p\}$. Wlog we can assume that V is C -monotonic, otherwise we take the C -monotonic cover of V instead of V . We construct I as follows

$$I_1 = v(\alpha_1)$$

$$I_{k+1} = v(\bigcup \{\alpha_1, \dots, \alpha_{k+1}\}) - v(\bigcup \{\alpha_1, \dots, \alpha_k\}) \quad 1 \leq k \leq p-1$$

and for every $T \in C$ we put

$$I(T) = \sum_{\substack{i \in \{1, \dots, p\} \\ T \supset \alpha_i}} I_i$$

That I has the desired property is a direct consequence of Theorem 4 of Shapley (1971). The conclusion follows from Theorem 3.4 and from the inclusion $C(I, R) \subset C(V, R) \forall R \in \mathbb{R}^N$. The second part of the theorem is straightforward.

3.7 Concluding Remark: Suppose that A is a finite set that μ is a probability measure on A ($M = P(A)$), and that V be the set of valuated VCs. Then conclusions of Proposition 2.8 do not hold in the "only if" part (see Abdou 1982, Example 3). This is the main consequence of the presence of "atoms" for the measure μ .

As a consequence of this remark, the continuous model presented in Section II does not generalize the discrete model of Section I. Although essential results on stability do hold in both contexts, we need a general theory which contains both of them.

Appendix

We begin this appendix by stating two lemmas which are direct corollaries of the Saks theorem (see Warga 1972).

Lemma 1: Let (A, M, μ) be a nonatomic probability measure space. Then for any couple (B_1, B_2) of elements of M such that $B_1 \subset B_2$, and any real number m such that $\mu(B_1) \leq m \leq \mu(B_2)$, there exists an element B of M such that

$$B_1 \subset B \subset B_2 \quad \text{and} \quad \mu(B) = m.$$

Lemma 2: The space (A, M, μ) is as in Lemma 1. For any finite collection $(m_i)_{i=1,2,\dots,r}$ of real numbers such that $m_i \geq 0$ ($i = 1, 2, \dots, r$) and $\sum_{i=1}^r m_i \leq 1$ there exists a collection $(B_i)_{i=1,2,\dots,r}$ of elements of M such that

$$\mu(B_i) = m_i \quad i = 1, 2, \dots, r$$

$$B_i \cap B_j = \emptyset \quad i, j = 1, 2, \dots, r \quad i \neq j.$$

Proof of Proposition 2.8: Let $V \in \mathcal{V}$ and let v be its associated VPF. Assume that v is not well-behaved from below; there exists a finite collection $(T_i)_{i=1,\dots,r}$ such that

$$T_i \in \mathcal{C} \quad i = 1, \dots, r$$

$$T_i \cap T_j = \emptyset \quad i, j = 1, \dots, r, \quad i \neq j, \quad \text{and}$$

$$\sum_{i=1}^{i=r} v(T_i) > 1.$$

Without loss of generality we assume $v(T_i) > 0 \quad \forall i = 1, \dots, r$.

There exists an integer $r' < r$ such that

$$v(T_1) + \dots + v(T_{r'}) \leq 1$$

and

$$v(T_1) + \dots + v(T_{r'+1}) > 1.$$

Put $\alpha = \frac{1}{r'+1} \left(\sum_{i=1}^{r'+1} v(T_i) - 1 \right) > 0$. For any ϵ satisfying $0 < \epsilon < \inf \left(\alpha, \frac{v(T_1)}{r'+1}, \dots, \frac{v(T_{r'+1})}{r'+1} \right)$ there exists, in view of Lemma 2, a collection of subsets of $A : (B_i), i = 1, \dots, r'$, such that

$$B_i \cap B_j = \emptyset; \quad i, j = 1, \dots, r'$$

$$\mu(B_i) = v(T_i) - \epsilon; \quad i = 1, \dots, r'$$

$$B_i \in \mathbb{M}; \quad i = 1, \dots, r'.$$

Put $\omega = A - \bigcup_{i=1}^{i=r'} B_i$; we have $\omega \in \mathbb{M}$ and

$$\begin{aligned} \mu(\omega) &= 1 - \sum_{i=1}^{r'} \mu(B_i) = 1 - \sum_{i=1}^{i=r'} v(T_i) + r' \epsilon \\ &= v(T_{r'+1}) - r'(\alpha - \epsilon) - \alpha < v(T_{r'+1}). \end{aligned}$$

If we put $B_{r'+1} = \omega$ we have

$$T_i \cap T_j = \emptyset \quad i = 1, 2 \dots r' + 1, \quad i \neq j$$

$$B_i \in V(T_i) \quad i = 1, 2 \dots r' + 1$$

$$\bigcup_{i=1}^{r'+1} B_i = A.$$

Then V is not well-behaved from below.

Assume now that v is not well-behaved from above: there exists a finite collection $(T_i) \ i = 1, \dots, r$, such that

$$T_i \in \mathcal{C} \quad i = 1, \dots, r,$$

$$\bigcup_{i=1}^r T_i = \emptyset, \quad \text{and}$$

$$\sum_{i=1}^{i=r} (1 - v(T_i)) < 1.$$

For every $i = 1, \dots, r$ we have $v(T_i) > 0$. Put

$$\alpha = \inf \left[\frac{1}{2r} \left(1 - \sum_{i=1}^{i=r} (1 - v(T_i)) \right), \frac{v(T_1)}{2}, \dots, \frac{v(T_r)}{2} \right]$$

For every $i = 1, \dots, r$ there exists a set $G_i \in V(T_i)$ such that $v(T_i) - \alpha < \mu(G_i) < v(T_i)$.

We have the inequalities

$$\sum_{i=1}^r (1 - v(T_i)) < \sum_{i=1}^r (1 - \mu(G_i)) < \sum_{i=1}^{i=r} (1 - v(T_i)) + r\alpha < 1.$$

In view of Lemma 2 there exists a finite collection $(F_i), i = 1, \dots, r$ of measurable sets such that

$$\mu(F_i) = 1 - \mu(G_i) \quad i = 1, \dots, r$$

$$F_i \cap F_j = \emptyset \quad i, j = 1, \dots, r, \quad i \neq j$$

If we put $B_i = A - F_i$ it entails that $\mu(B_i) = \mu(G_i)$ and that $B_i \in V(T_i), i = 1, \dots, r$ which means that V is not well-behaved from above.

Conversely assume that V is not well behaved from below: there exists a collection $(T_i, B_i), i = 1, \dots, r$ such that $B_i \in V(T_i), i = 1, \dots, r; T_i \cap T_j = \emptyset, i, j = 1, \dots, r, i \neq j; \bigcup_{i=1}^{i=r} B_i = A$; and $\sum_{i=1}^r v(T_i) > \sum_{i=1}^r \mu(B_i) \geq 1$ which means that v is not well-behaved from below.

Assume now that V is not well-behaved from above. There exists a collection $(T_i, B_i), i = 1, \dots, r$ such that

$$B_i \in V(T_i) \quad i = 1, \dots$$

$$B_i \cup B_j = A \quad i, j = 1, 2, \dots, r, \quad i \neq j$$

$$\bigcap_{i=1}^r T_i = \emptyset.$$

We have $(A - B_i) \cap (A - B_j) = \emptyset, \forall i, j = 1, \dots, r, i \neq j$ and

$$\sum_{i=1}^{i=r} (1 - v(T_i)) < \sum_{i=1}^{i=r} (1 - \mu(B_i)) = \sum_{i=1}^r \mu(A - B_i) \leq 1$$

which means that v is not well-behaved from above.

As the complete proofs of all statements contained in the proposition are similar (and tedious), we give an extended proof only for convexity. The reader can easily provide proofs for the remainder.

Let $V \in \mathcal{V}$ and assume that v is convex. Take T_1 and T_2 in $\mathcal{C}$, B_1 and B_2 in $\mathcal{M}$ such that $B_1 \in V(T_1)$ and $B_2 \in V(T_2)$.

If we had $B_1 \cap B_2 \notin V(T_1 \cap T_2)$ and $B_1 \cup B_2 \notin V(T_1 \cup T_2)$, we should derive the inequalities $v(T_1) + v(T_2) > \mu(B_1) + \mu(B_2) = \mu(B_1 \cup B_2) + \mu(B_1 \cap B_2) \geq v(T_1 \cup T_2) + v(T_1 \cap T_2)$ which would contradict the convexity of v .

Now we want to prove that if V is in $\mathcal{V}^*$ or in $\mathcal{V}_c$ and V is convex then v is convex. First we notice that if V is in $\mathcal{V}^*$ and V is convex then V is in $\mathcal{V}_c$ too, so that we can restrict the proof to $\mathcal{C}$ -monotimic VCs. We seek to prove

$$\forall T_1 \in \mathcal{C}, \quad \forall T_2 \in \mathcal{C} \quad v(T_1) + v(T_2) \leq v(T_1 \cup T_2) + v(T_1 \cap T_2).$$

If $v(T_1) = 0$ or $v(T_2) = 0$, then by the $\mathcal{C}$ -monotonicity of V the inequality is obviously true.

Suppose now $v(T_1) > 0$ and $v(T_2) > 0$ and assume, for the sake of a contradiction, that we have

$$v(T_1) + v(T_2) = v(T_1 \cup T_2) + v(T_1 \cap T_2) + \alpha, \quad (1)$$

with $\alpha > 0$. In view of the $\mathcal{C}$ -monotonicity of V we have

$$v(T_1) \leq v(T_1 \cup T_2)$$

$$v(T_2) \leq v(T_1 \cup T_2)$$

$$v(T_1 \cap T_2) \leq v(T_1)$$

$$v(T_1 \cap T_2) \leq v(T_2).$$

It is impossible for either of the last two inequalities to be an equality, otherwise the two first inequalities would contradict (1). Hence there exist an $\epsilon > 0$ and an $\eta > 0$ satisfying

$$\epsilon < \inf (\alpha, v(T_1) - v(T_1 \cap T_2), v(T_2) - v(T_1 \cap T_2))$$

and $\eta + \epsilon < \alpha$.

Since $0 \leq v(T_1 \cap T_2) \leq 1$, by applying Lemma 1, there exists a measurable set I such that $\mu(I) = v(T_1 \cap T_2)$. Since $0 \leq v(T_1 \cap T_2) < v(T_1) - \epsilon \leq 1$ by applying Lemma 1 again, there exists $B_1 \in \mathcal{M}$ such that $I \subset B_1$ and $\mu(B_1) = v(T_1) - \epsilon$. Put $m =$

$\inf (v(T_1 \cup T_2) + \alpha - (\eta + \epsilon), 1)$. Since

$$0 \leq v(T_1 \cap T_2) < v(T_1) - \epsilon < v(T_1 \cup T_2) < v(T_1 \cup T_2) + \alpha - (\eta + \epsilon)$$

we have $0 \leq v(T_1) - \epsilon < m \leq 1$ and by applying Lemma 1, there exists $G \in \mathbb{M}$ such that $B_1 \subset G$ and $\mu(G) = m$.

Now we put $B_2 = I \cup (G \setminus B_1)$ (disjoint union). We have

$$\mu(B_2) = \mu(I) + \mu(G - B_1) \leq v(T_1 \cap T_2) + [v(T_1 \cup T_2) + \alpha - \eta - \epsilon]$$

$$- [v(T_1) - \epsilon] = v(T_2) - \eta$$

It is clear that $B_1 \in V(T_1)$, $B_2 \in V(T_2)$ and $B_1 \cup B_2 = G \notin V(T_1 \cup T_2)$, and $B_1 \cap B_2 = I \notin V(T_1 \cap T_2)$. Q.E.D.

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